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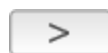
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PatternToolDefTopic



About PatternTool

About PatternTool

PatternTool allows you to:

- [Create, edit, and debug pattern files](#)
- [Overlay failures on programmed vectors](#)

The following topics are included:

- [Opening and Managing Pattern Files](#)
- [Viewing Pattern Data](#)
- [Editing Pattern Data](#)
- [Using the Explorer and Properties Windows](#)
- [Bursting Patterns](#)
- [PatternTool Toolbar and Command Reference](#)

IG-XL includes additional digital tools:

- [Pattern Control Display:](#)
 - [Displays pattern information and memory consumption](#)
 - [Provides interactive pattern execution and management](#)
- [HRAM Display:](#)
 - [Provides a tabular view of captured HRAM data](#)
 - [Includes fail overlay and actuals](#)

- **Waveform Display:**

- Graphically view waveforms over time and across pins of interest
- Supports Protocol Aware (PA) frames and standard digital
- Performs actual mode

- **Protocol Definition Editor:**

- Define a protocol from scratch
- Edit an existing protocol definition
- Graphically view frame waveforms

- **Protocol Studio:**

- Displays Protocol Aware (PA) frames at various levels of abstraction
- Enables programming of PA port settings
- Enables initiation of frames

- **CyclePro:**

- Raw cycle display mode
- Symbol10b display mode
- PCIe Link traffic decode (CMEM) and ATE Tx traffic decode (Tx HRAM)
- Layer Selection: Transaction, Data Link, PHY

For more information, see:

- [Pattern Control Display](#)
- [About HRAM Display](#)
- [About Waveform Display](#)
- [About Protocol Definition Editor](#)
- [About Protocol Studio](#)
- [About CyclePro](#)

For more information on patterns, see:

- [Pattern Compiler](#)
- [Pattern Reverse Compiler](#)

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Bursting Patterns

Bursting Patterns

This section provides information on using PatternTool to burst patterns. The following topics are available:

- [Bursting a Pattern](#)
- [Using Learn Mode](#)
- [Running Actual Mode](#)

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Bursting Patterns : Bursting a Pattern

Bursting a Pattern

PatternTool lets you burst a pattern module, file, or set.

The following topics are included:

- [Loading and Unloading Patterns](#)
- [Burst Options](#)
- [Setting HRAM Capture](#)
- [Accumulate Fails](#)
- [Masking Pins and Vectors](#)
- [Updating the Tester](#)

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Bursting Patterns : Bursting a Pattern : Loading and Unloading Patterns

Loading and Unloading Patterns

A pattern file must be loaded into memory before you can burst it. Validating a program loads all the patterns it uses. You do not need to load them again. If for some reason you need to load or unload a pattern, select it in the Pattern Explorer and select Debug>Load/Unload Pattern, or use the command from the right-click context menu.

If a module is loaded as part of a pattern file or pattern set, you cannot unload it individually. The Unload command, therefore, is unavailable for such a file or module.

You can use Load if you have opened a file using File>Open but have not loaded it into tester memory. You can load the file to burst it. The command loads the current pattern file, not the current module. If the pattern has been modified, Load performs pattern compilation or tester update so that the loaded pattern reflects the pattern displayed in the tool.

If you have made edits that you now want to cancel, instead of using Undo you may find it easier to reload the pattern from disk into PatternTool. This restores the pattern to its last saved state. Use File>Reload from Saved Pattern or Ctrl+Shift+O. The command also reloads the pattern to the tester. You cannot undo this command.

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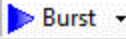
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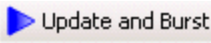


Bursting Patterns : Bursting a Pattern : Burst Options

Burst Options

The buttons and commands for bursting the current pattern are as follows. Burst is the default option for the Burst button. To access the menu of other options, click the triangle portion of the button.

Burst	 Burst Ctrl+B
	Bursts the currently displayed pattern once.
Burst Loop	 Burst Loop
	Bursts the currently displayed pattern in a loop. The status bar Burst Count indicates the number of times the pattern has been burst. You can use Halt to stop the loop.
Burst Until Pass	 Burst Until Pass
	Bursts the currently displayed pattern in a loop until the pattern passes. The status bar Burst Count indicates the number of times the pattern has been burst. You can use Halt to stop the loop.
Burst Until Fail	 Burst Until Fail Ctrl+Shift+B
	Bursts the currently displayed pattern in a loop until the pattern fails. The status bar Burst Count indicates the number of times the pattern has been burst. You can use Halt to stop the loop.
Accumulate Fails	 Accumulate Fails
	Repeatedly bursts the currently displayed pattern to accumulate fails from the entire pattern. See Accumulate Fails.

 **Update and Burst** If a pattern has been modified but the tester has not been updated, the burst commands and button prepend the text Update and. If edits are structural, you must recompile the pattern before you can update it to the tester. Structural edits include adding, renaming, or deleting pins or modules, or adding instrument microcode or PatGen opcodes (except repeat). Update and commands update the tester, if necessary, and execute the pattern file or pattern set that is currently open in the tool. These commands then display failing vectors and pins, if any. The result of the pattern burst is reported in the status bar.

You must load a pattern into tester memory before it can be burst. See Loading and Unloading Patterns.

By default, HRAM capture starts at vector 0. See Setting HRAM Capture.

 **Halt** During pattern execution, the Halt command is enabled. You can use this button, Debug>Halt, or Ctrl+T.

A test instance that bursts a pattern as part of a job may specify an interpose function to be executed before the pattern burst (PrePatF) or after the pattern burst (PostPatF). When you burst these patterns in PatternTool, you may want to execute one or both of these interpose functions as well. PatternTool lets you enable or disable execution of one or both functions:

Debug>PrePatF	 PrePatF
	Enable or disable the execution of the PrePatF function before each pattern burst.
Debug>PostPatF	 PostPatF
	Enable or disable the execution of the PostPatF function after each pattern burst.

Note that these settings are uniform across PatternTool, Waveform Display, and HRAM Display. Changing a setting in one tool changes it for the others as well. In this way, pattern reburst is the same across all tools.

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Bursting Patterns : [Bursting a Pattern](#) : Setting HRAM Capture

Setting HRAM Capture

[PatternTool](#) lets you set the following aspects of HRAM capture:

- [Setting What To Capture](#)
- [Setting Start and Stop](#)

Setting What To Capture

You can specify what is to be captured on the next burst. The Capture control is on the toolbar:

Capture: Fails ▼

Capture Control

You can also use `Debug>HRAM Capture` command.

The options are:

- [Fails](#)
- [STV](#)
- [FirstModFail](#)

Setting Start and Stop

By default, when you reburst a pattern, HRAM capture starts at vector 0. It continues until HRAM is full. If vectors that interest you come later in the pattern, HRAM may not be captured for those vectors. You can therefore set capture to start at a later point and to stop after vectors of interest.

To set capture start:

1. Select the vector where you want capture to start.
2. Right-click, and select Set/Reset Capture Start from the context menu. You can also use Ctrl+Q. The command is also on the Debug menu.

Right-double-clicking in the leftmost column of a vector also sets, moves, or removes the start capture point at that vector.

A green right-pointing triangle appears in the leftmost column to mark the starting vector. A vertical green line appears in the viewstrip.

To set capture stop:

1. Select the vector where you want capture to stop.
2. Right-click, and select Set/Reset Capture Stop from the context menu. The command is also on the Debug menu.

A light blue right-pointing triangle appears in the leftmost column to mark the starting vector. A vertical light blue line appears in the viewstrip.

The following figure shows captured fail data using the default start capture, vector 0, and the default stop:

TSet	Command	Label	Vector	P 0	P 1	P 2	P 3	P 4	PPPPPP1 567890	PinGroup_11	PinGroup_21	PinGroup_31to50	PinGroup_51to65
ts0			0	L	L	L	L	L	LLLLLL	LLLLLLLLLLLL	LLLLLLLLLLLLLLLLLLLL	LLLLLLLLLLLLLLLLLLLLLLLLLLLL	LLLLLLLLLLLLLLLLLLLL
ts0			1	X	0	0	X	X	000000	XXXXXXXXXX	0000000000	000000000000000000000000	XXXXXXXXXXXXXXXXXX
ts0	repeat...		2	H	0	0	X	0	00 000	XXXXXXXXXX	0000000000	000000000000000000000000	XXXXXXXXXXXXXXXXXX
ts0			3	0	0	0	H	0	000000	0H0H0H0H0H	0000000000	000000000000000000000000	LHLHLHLHLHLHLHL
ts0			4	0	0	0	H	0	000000	0H0H0H0H0H	0000000000	000000000000000000000000	LHLHLHLHLHLHLHL
ts0			5	0	0	0	H	0	000000	0H0H0H0H0H	0000000000	000000000000000000000000	LHLHLHLHLHLHLHL
ts0			6	0	0	0	H	0	000000	0H0H0H0H0H	0000000000	000000000000000000000000	LHLHLHLHLHLHLHL
ts0			7	X	0	0	H	0	000000	0H0H0H0H0H	0000000000	000000000000000000000000	LHLHLHLHLHLHLHL
ts0			8	0	0	0	H	0	000000	0H0H0H0H0H	0000000000	000000000000000000000000	LHLHLHLHLHLHLHL
ts0			9	0	H	0	H	0	000000	0H0H0H0H0H	0000000000	000000000000000000000000	LHLHLHLHLHLHLHL
ts0			10	0	0	0	H	0	000000	0H0H0H0H0H	0000000000	000000000000000000000000	LHLHLHLHLHLHLHL
ts0			11	0	0	0	H	0	000000	0H0H0H0H0H	0000000000	000000000000000000000000	LHLHLHLHLHLHLHL
ts0			12	0	H	0	H	0	000000	0H0H0H0H0H	0000000000	000000000000000000000000	LHLHLHLHLHLHLHL
ts0			13	0	0	0	H	0	000000	0H0H0H0H0H	0000000000	000000000000000000000000	LHLHLHLHLHLHLHL
ts0			14	0	0	0	H	0	000000	0H0H0H0H0H	0000000000	000000000000000000000000	LHLHLHLHLHLHLHL
ts0			15	0	0	0	H	0	000000	0H0H0H0H0H	0000000000	000000000000000000000000	LHLHLHLHLHLHLHL

Example of Captured Fail Data Starting at Vector 0

In the following figure, the start point is set on the vector after the previous failing vector and the stop point is set a few vectors later.

[illegible]

Example of Captured Fail Data With Nondefault Start and Stop

Note that the Accumulate Fails command repeatedly bursts a pattern until all fails in the pattern are captured. See [Accumulate Fails](#).

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Bursting Patterns : Bursting a Pattern : Accumulate Fails

Accumulate Fails

By default, bursting a pattern captures failures in HRAM starting at vector 0 until HRAM is full. To get fails for an entire pattern, you can use the Accumulate Fails feature.



Accumulate Fails

Use this button or Debug>Accumulate Fails.

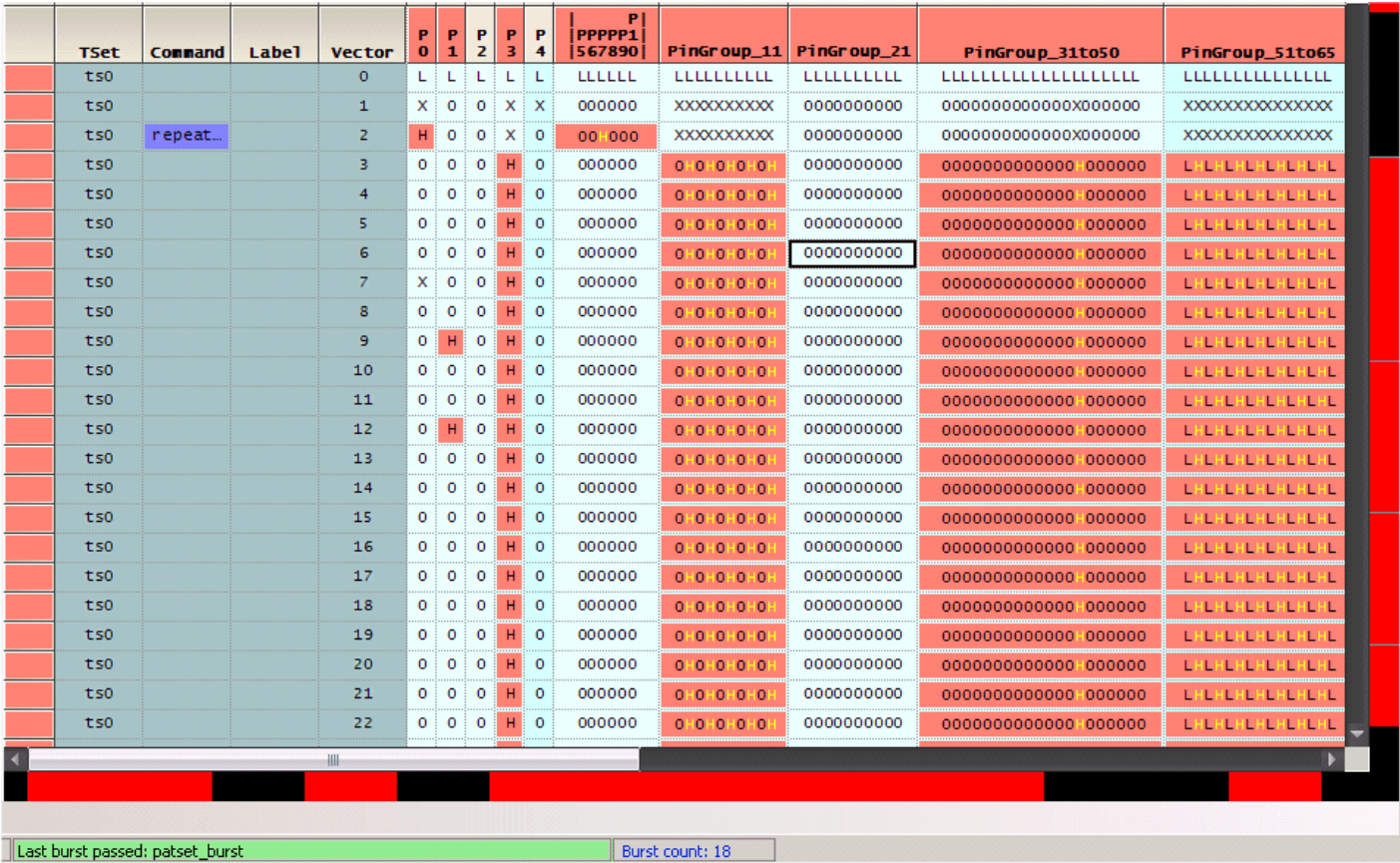
This command bursts a pattern starting at vector 0 and runs until HRAM is full. It then moves the start capture point to the vector where the fail occurred and rebursts. It continues this process until it reaches the end of the pattern. The Burst count value in the status bar indicates how many bursts were required to accumulate all fails.

The following figure shows fails captured by a regular Burst starting at vector 0:

TSet	Command	Label	Vector	P 0	P 1	P 2	P 3	P 4	PPPPPP1 567890	PinGroup_11	PinGroup_21	PinGroup_31to50	PinGroup_51to65
ts0			0	L	L	L	L	L	LLLLLL	LLLLLLLLLL	LLLLLLLLLL	LLLLLLLLLLLLLLLLLLLL	LLLLLLLLLLLLLLLL
ts0			1	X	0	0	X	X	000000	XXXXXXXXXX	0000000000	000000000000X000000	XXXXXXXXXXXXXXXXXX
ts0	repeat...		2	H	0	0	X	0	00 000	XXXXXXXXXX	0000000000	000000000000X000000	XXXXXXXXXXXXXXXXXX
ts0			3	0	0	0	H	0	000000	0H0H0H0H0H	0000000000	000000000000H000000	LHLHLHLHLHLHL
ts0			4	0	0	0	H	0	000000	0H0H0H0H0H	0000000000	000000000000H000000	LHLHLHLHLHLHL
ts0			5	0	0	0	H	0	000000	0H0H0H0H0H	0000000000	000000000000H000000	LHLHLHLHLHLHL
ts0			6	0	0	0	H	0	000000	0H0H0H0H0H	0000000000	000000000000H000000	LHLHLHLHLHLHL
ts0			7	X	0	0	H	0	000000	0H0H0H0H0H	0000000000	000000000000H000000	LHLHLHLHLHLHL
ts0			8	0	0	0	H	0	000000	0H0H0H0H0H	0000000000	000000000000H000000	LHLHLHLHLHLHL
ts0			9	0	H	0	H	0	000000	0H0H0H0H0H	0000000000	000000000000H000000	LHLHLHLHLHLHL
ts0			10	0	0	0	H	0	000000	0H0H0H0H0H	0000000000	000000000000H000000	LHLHLHLHLHLHL
ts0			11	0	0	0	H	0	000000	0H0H0H0H0H	0000000000	000000000000H000000	LHLHLHLHLHLHL
ts0			12	0	H	0	H	0	000000	0H0H0H0H0H	0000000000	000000000000H000000	LHLHLHLHLHLHL
ts0			13	0	0	0	H	0	000000	0H0H0H0H0H	0000000000	000000000000H000000	LHLHLHLHLHLHL
ts0			14	0	0	0	H	0	000000	0H0H0H0H0H	0000000000	000000000000H000000	LHLHLHLHLHLHL
ts0			15	0	0	0	H	0	000000	0H0H0H0H0H	0000000000	000000000000H000000	LHLHLHLHLHLHL
ts0			16	0	0	0	H	0	000000	0H0H0H0H0H	0000000000	000000000000H000000	LHLHLHLHLHLHL
ts0			17	0	0	0	H	0	000000	0H0H0H0H0H	0000000000	000000000000H000000	LHLHLHLHLHLHL
ts0			18	0	0	0	H	0	000000	0H0H0H0H0H	0000000000	000000000000H000000	LHLHLHLHLHLHL
ts0			19	0	0	0	H	0	000000	0H0H0H0H0H	0000000000	000000000000H000000	LHLHLHLHLHLHL
ts0			20	0	0	0	H	0	000000	0H0H0H0H0H	0000000000	000000000000H000000	LHLHLHLHLHLHL
ts0			21	0	0	0	H	0	000000	0H0H0H0H0H	0000000000	000000000000H000000	LHLHLHLHLHLHL
ts0			22	0	0	0	H	0	000000	0H0H0H0H0H	0000000000	000000000000H000000	LHLHLHLHLHLHL

Captured Fail Data Starting at Vector 0

The following figure shows the result of the Accumulate Fails command. The Burst count in the status bar indicates that 18 bursts were required to accumulate all fails.



Accumulated Fail Data Starting at Vector 0

You can also set a nondefault start of capture to get fails for a limited set of vectors. See [Setting HRAM Capture](#).

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Bursting Patterns : Bursting a Pattern : Masking Pins and Vectors

Masking Pins and Vectors

Masking is a common debug activity that lets you concentrate on a subset of failures of interest. For example, during silicon bring-up, a device might have a faulty pin that fails on every vector. You can mask failures on this pin, allowing the test system to capture and display failure information on other pins.

To mask a pin:

1. Select the pin column header in the main display or select the pin in the Pin Explorer.
2. Select Debug>Mask, or Ctrl+Shift+K, or right-click and select Mask from the context menu. If you select a masked pin, the command becomes Unmask. The same control sequence unmaskes a masked pin.

When you mask a pin, the header of the pin column is shown in blue. In addition, the pin in the Pin Explorer is shown in blue:

Pin Explorer

Include these pins:

All Pins

Failing Pins

Show selected pins:

☒ Pins

TSet

Command

Label

Vector

P0

P1

P2

P3

P4

	TSet	Command	Label	Vector	P0	P1	P2	P3	P4	P	PinGroup_11
	ts0			0	L	L	L	L	L	LLLLLL	LLLLLLLLLL
	ts0			1	X	0	0	X	X	000000	XXXXXXXXXX
	ts0	repeat...		2	H	0	0	X	0	00,000	XXXXXXXXXX
	ts0			3	0	0	0	H	0	000000	0H0H0H0H
	ts0			4	0	0	0	H	0	000000	0H0H0H0H
	ts0			5	0	0	0	H	0	000000	0H0H0H0H
	ts0			6	0	0	0	H	0	000000	0H0H0H0H
	ts0			7	X	0	0	H	0	000000	0H0H0H0H
	ts0			8	0	0	0	H	0	000000	0H0H0H0H
	ts0			9	0	H	0	H	0	000000	0H0H0H0H
	ts0			10	0	0	0	H	0	000000	0H0H0H0H

Pin Data and Pin Explorer Showing a Masked Pin (P0)

If you select a range of pins that includes a masked pin, the Unmask command is enabled. You must unmask the masked pin before you can mask the entire range of pins.

You mask and unmask pins by disabling and enabling the pin comparator. The command therefore applies only to digital pins. You can also mask a pin by changing all its data to X, but using the Mask and Unmask commands is faster.

Using this feature on a pin that is currently permanently masked has no effect on the pin. The VBT statement to mask pins is `TheHdw.Digital.Pins.DisableCompare`.

Masking vectors lets you investigate failures by enabling and disabling failures on a per-vector basis. For example, you can mask a selected range of vectors across all pins. Masking a vector lets you ignore the results on selected vectors of choice at one time versus disabling each vector/cell combination one at a time (by changing the data to X).

You mask a vector by editing its Command cell and adding the mask control bit, which is in the Control Bits section of the command edit window. See [Editing Command Cells](#). When you are finished debugging, remove the mask control bits. Otherwise, they are saved to the pattern file.

		
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Bursting Patterns : Bursting a Pattern : Updating the Tester

Updating the Tester

One use of PatternTool is to edit a binary pattern file to run it immediately on the tester to try out the changes. PatternTool automatically makes this part of bursting a pattern. If you have made edits to the current pattern, the Burst commands change to Update and Burst. You therefore do not need to update the tester explicitly before rebursting.

 **Update Tester** If you want to update the tester without bursting, use this button, the Debug>Update Tester command, or Ctrl+U.

Some changes to a pattern cannot simply be updated to the tester. These include major structural changes, such as adding PatGen commands or adding or deleting pins. These changes require you to reload a module before the tester can be updated. You can use the command Debug>Update Tester (Force Reload) or Ctrl+Shift+U to perform this complete reload and update operation. This may be useful if the tester memory is out of date in respect to the currently opened pattern.

	
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Bursting Patterns : Using Learn Mode

Using Learn Mode

Learn Mode lets you use results captured in HRAM to correct a failing pattern. You can use Learn Mode to update a failing pattern, either by changing the expect data to match the values retrieved from HRAM so that the expect data will match the comparator data, or by changing the expect data to X so that failures are ignored.

The major steps in using Learn Mode are:

- Specify Learn Mode settings that determine how PatternTool is to correct the pattern
- Optionally specify that a Learn Mode report be generated
- Optionally select the portion of the pattern that Learn Mode is to act on
- Run Learn Mode to correct the pattern

The following topics are available:

- Learn Mode Settings
- Executing Learn Mode
- Learn Mode Report

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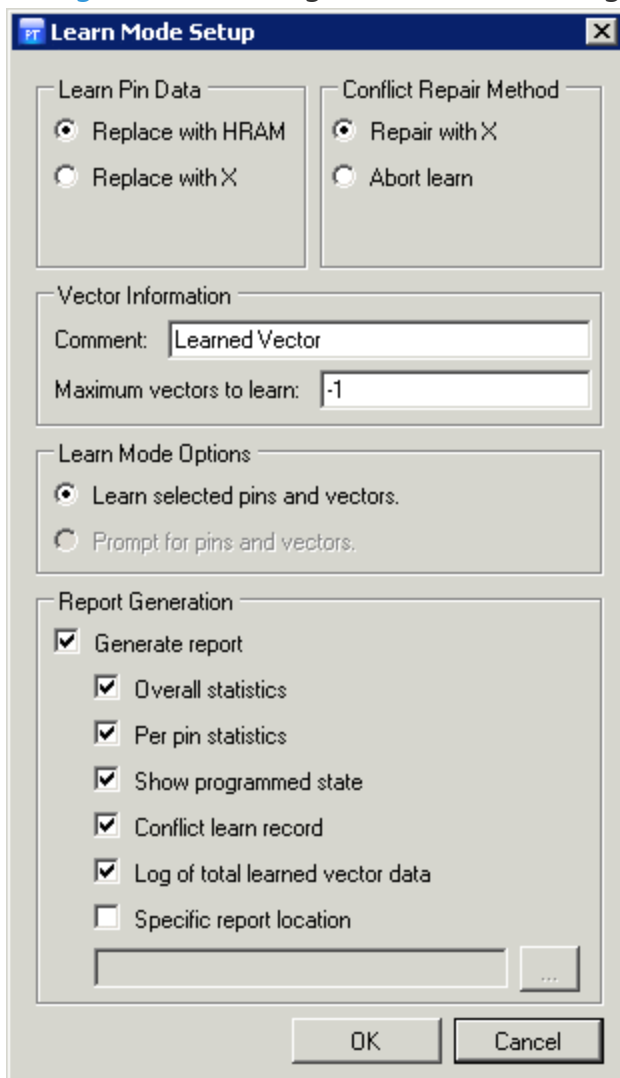
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Bursting Patterns : Using Learn Mode : Learn Mode Settings

Learn Mode Settings

You can specify settings that determine how Learn Mode will correct a pattern. To bring up the dialog box, use [Debug>Learn Mode Settings](#) or [Ctrl+D](#). The dialog box looks like this:



The **Learn Mode Setup** dialog box is used to configure settings for the Learn Mode. It contains the following sections:

- Learn Pin Data**:
 - ☒ Replace with HRAM
 - ☐ Replace with X
- Conflict Repair Method**:
 - ☒ Repair with X
 - ☐ Abort learn
- Vector Information**:
 - Comment:
 - Maximum vectors to learn:
- Learn Mode Options**:
 - ☒ Learn selected pins and vectors.
 - ☐ Prompt for pins and vectors.
- Report Generation**:
 - ☒ Generate report
 - ☒ Overall statistics
 - ☒ Per pin statistics
 - ☒ Show programmed state
 - ☒ Conflict learn record
 - ☒ Log of total learned vector data
 - ☐ Specific report location

At the bottom, there is a text field and a button (likely "Browse...") for the report location, and **OK** and **Cancel** buttons.

Learn Mode Setup Window

Settings persist for the current PatternTool session.

The parts of the Learn Mode Settings window are described in the following sections:

- Learn Pin Data
- Conflict Repair Method
- Vector Information
- Learn Mode Options
- Report Generation

Learn Pin Data

This option specifies what should replace the failing data. The options are:

Replace with HRAM	Replace the expect data with the data retrieved from HRAM to repair the failures. This involves swapping L and H or swapping M and V. Default.
Replace with X	Replace the expect data with X (Don't care) so that the comparator will ignore the failure and the vector will pass.

Conflict Repair Method

In some cases, Learn Mode cannot repair a failing vector because the value that it substitutes does not cause the pattern to pass. This is known as a conflict. The pattern may exhibit intermittent behavior so that it does not consistently pass or fail. For example, if there is a loop and the HRAM data changes for each cycle, a conflict occurs.

You must specify what Learn Mode is to do when it detects a conflict:

Repair with X	Replace the conflict data with X so that the comparator will ignore the failure. This is the default. Conflicts are represented in the display as a light-blue crosshatch over the vector or pin with the conflict.
Abort learn	Abort Learn Mode. Note that this may leave the pattern partially learned.

Vector Information

This section has the following fields:

Comment	Specifies a comment to be inserted on each vector that has been learned. The default comment is "Learned Vector." If a learned vector already has a comment, the Learn Mode comment is appended to it.
Maximum vectors to learn	Specifies a maximum number of vectors to be learned, after which PatternTool will abort Learn Mode. Learn Mode can take a long time, depending on the number of vectors to be learned. This option lets you specify a maximum. The default is -1, which means to continue to learn until all selected vectors are processed. If Learn Mode does abort because it has reached the specified maximum, vectors that have already been learned remain learned.

Learn Mode Options

This section specifies which pins and vectors are to be learned. The options are:

Learn selected pins and vectors	You manually select the pins and vectors to be learned in the pattern display before executing Learn Mode. This option is always selected.
Prompt for pins and vectors	Not supported.

Report Generation

You can request a report on the results of a Learn Mode operation. You can also specify what information you want in the report.

Check Generate report to get a report. Check the other boxes for specific information:

• Overall statistics

• Per pin statistics

• Show programmed state

• Conflict learn record

• Log of total learned vector data

By default, all these boxes are checked.

An additional check box, Specific report location, is not checked by default. If you check it, you can enter the location you want for the report, or you can click the button to browse to it.

For more information, see [Learn Mode Report](#).

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Bursting Patterns : Using Learn Mode : Executing Learn Mode

Executing Learn Mode

Before running Learn Mode, you can select the pins and vectors on which Learn Mode is to operate. By default, all pins and vectors are selected.



To execute Learn Mode, click this button, use Ctrl+L, or select Debug>Learn....

Once Learn Mode starts, its operation is automatic. PatternTool repeatedly bursts the pattern to capture failure information, and it uses the information to modify the pattern. It automatically updates the tester and performs a burst to verify the changes. Failures to fix are treated as conflicts. If there are more failures than can fit in HRAM, it performs additional bursts as needed until it has corrected all failures in the selected area.

Note the following points about Learn Mode:

- Learn Mode supports parallel vectors and scan vectors.
- Learn Mode is applied only to digital pins. Learn Mode ignores any analog pins that are included in the selection. Learn Mode does not make any visual changes to the display of selected analog pins.
- If Learn Mode encounters an instrument that it cannot learn, it considers that pin "irreparable" and marks it as a conflict.
- Masked pins and vectors are not considered during learning because Learn Mode applies only to failed pins and vectors.

- Learn Mode operates only on the currently selected site. All other sites are disabled during learning and then restored to their previous state once learning is complete.
- Learn Mode operates only on modules currently opened in PatternTool. For example, a pattern set can include modules from different pattern files. Learn Mode considers only modules in the currently displayed pattern set, not other modules in the pattern files.
- Learn Mode cannot be run for Keep Alive.
- Learn Mode is restricted to one million cells. If you try to learn more than one million cells, a warning appears.

Learning... Cancel While Learn is executing, this appears message in the status bar. Click Cancel to cancel the Learn operation. The pattern remains in whatever state it is in when you halt the operation. This means that the pattern may be partially learned.

If you do not want to save the pattern with the completely or partially learned data, you can simply reload the pattern into tester memory. Because you have not explicitly saved the learned changes, they are discarded.

When a cell value has been successfully learned, the cell is outlined in blue to indicate that it has been modified:

X	0	0	0	0	0	0	0
0	0	0	X	0	0	0	0
0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0

Pin Data Showing a Learned Cell

If the learned pin is part of a bus, only the pin cell is outlined.

In the Learn Mode Settings box, you can specify text that is to be written as a comment to all learned vectors. By default, the comment is "Learned Vector."

If Learn Mode cannot correct a failure, it treats the uncorrectable data as a conflict. Its action in case of conflict depends on the setting specified on the Learn Mode Setting box. Depending on the setting, it can change the cell value to X or it can abort learning. Learn Mode marks conflict cells with light blue cross-hatching.

For more information, see Conflict Repair Method.

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Bursting Patterns : Using Learn Mode : Learn Mode Report

Learn Mode Report

The Report Generation section of the Learn Mode Setup window lets you specify whether a Learn Mode report is to be generated and what information it should contain.

A separate report is created for each pattern file. By default, the report is created in the same folder as the pattern file, if possible. If the report cannot be created in this folder, you are prompted for a location. You can also check the box Specific report location on the Learn Settings window to specify a nondefault location.

The report filename is as follows:

patternname_ChangeLog_timestamp.txt

The Learn Mode Setup window includes the following options under Report Generation:

Generate report	Generates a report. If this box is checked, the following check boxes are enabled.
Overall statistics	The report includes overall information on the learning: how many corrections were made and which vectors and cycles were learned.
Per pin statistics	The report includes information on a per-pin basis including: - Count: Number of learns performed on the pin. - First: The starting vector. - Last: The ending vector.

- **Conflicts:** The number of conflicts encountered.

If a pin displays all asterisks, it means that the pin could not be learned or repaired because learning is not defined for instruments of the type connected to the pin.

Show programmed state	Includes the programmed state of the pattern file in the report.															
Conflict learn record	The report lists in detail conflicts that were encountered while learning. It lists the vector, the reason for the conflict, the associated cycle, the pin, and the condition, if any, associated with the vector. For example: <table><tr><th>Vector</th><th>Reason</th><th>Cycle</th><th>Pin</th><th>Condition</th></tr><tr><td colspan="5">-----</td></tr><tr><td>25</td><td>Prev passing cycle</td><td>25</td><td>Pin2</td><td>Loop=5</td></tr></table>	Vector	Reason	Cycle	Pin	Condition	-----					25	Prev passing cycle	25	Pin2	Loop=5
Vector	Reason	Cycle	Pin	Condition												

25	Prev passing cycle	25	Pin2	Loop=5												
Log of total learned vector data	The report contains a full printout of the learned data for each vector. If the data has been modified, the new value is listed. A hyphen indicates no change. For example: <pre>LEARNEDPINS PPPPP IIIII NNNNN 12345 Learned Vector(s) 10 X-LH ...</pre> An exclamation point (!) represents a conflict. An asterisk (*) represents irreparable data.															

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Bursting Patterns : Running Actual Mode

Running Actual Mode

You can select a range of cells in the PatternTool display and run Actual Mode on them. Select a range of cells. Then select Tools>Actual Mode, which is also available on the context menu. This brings up Waveform Display, which executes and displays actual mode. For more information, see Using Actual Mode in the Waveform Display section.

	
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Editing Pattern Data

Editing Pattern Data

This section provides information on using PatternTool to edit a pattern. The following topics are available:

- Undoing and Redoing Edits
- Directly Editing Cells
- Cutting, Copying, and Pasting Cells
- Editing TSet Cells
- Editing Command Cells
- Editing Label Cells
- Editing Digital Pin Data
- Viewing and Editing Block Comments
- Inserting, Copying, and Deleting Vectors
- Using Find and Replace

- [Masking Transitions](#)
-
- [Viewing and Editing Scan Chains](#)
-
- [Editing Instrument Microcode](#)

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Editing Pattern Data : Undoing and Redoing Edits

Undoing and Redoing Edits

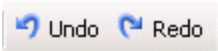
PatternTool lets you undo edits. You can also redo edits you have undone.

Undo and Redo operate on a single edit or on multiple edits:

- If the edit was to a single cell, the command undoes or redoes that single edit. This applies to a direct cell edit or to an edit made using an edit window.
- If the edit affected multiple cells through a single command, the command undoes or redoes the edits to all cells. For example, you can select a range of cells and then use the Complement command to switch each cell value to its complement. If you complement multiple cells in a single command, Undo undoes all the complements in a single operation. See [Complementing and Swapping Values](#).

[Commands on the](#) Edit menu explicitly indicate whether the next operation would be on a single cell or on multiple cells:

- [Undo/Redo](#) single cell edit - if a single cell was edited.
- [Undo/Redo](#) multiple cell edits - if multiple cells were edited using Complement or Swap.
- [Undo/Redo shift](#) - if a Shift modified multiple cells.
- [Undo/Redo paste](#) - if a Paste was the previous operation. The command does not indicate whether this was a single-cell or multiple-cell paste.



The toolbar buttons do not indicate whether the next operation would be for single or multiple edits, but they undo or redo single or multiple edits, depending on what the previous edit was.

You can use the standard control sequences Ctrl+Z for Undo and Ctrl+Y for Redo.

Note that you cannot undo Replicate commands. See [Replicating Values](#).

The Undo stack is cleared when you save the file or open a new file, or when you perform an undoable action such as inserting or deleting a module.

If you have made a number of edits that you now want to cancel, instead of using Undo it may be easier to reload the pattern from disk into PatternTool, which restores the pattern to its last saved state. Use File>Reload from Saved Pattern, or Ctrl+Shift+O. The command also reloads the pattern to the tester. You cannot undo this command.

		
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Editing Pattern Data : Directly Editing Cells

Directly Editing Cells

You can directly edit any PatternTool cell (except column headers cells and the vector number cell). Select a cell and directly type in it.

PatternTool provides editing tools for its non-data cells: Tset, Command, and Label, as well as instrument microcode. It is usually easier to use these editing tools. The menus will contain only valid values, and, for more complicated data, the dialog boxes can guide you through the valid options.

You can directly edit the pin data cells. You can also select a range of cells and then cut, copy, and paste their data, or you can use commands that shift, swap, complement, or replicate their data values.

You cannot enter an invalid value. When you enter an invalid value and then try to leave the cell, PatternTool does not let you leave the cell. It displays a message in its status bar and in its Message Display Panel saying that it cannot set the pin data or other value. You cannot leave until you enter a valid value or use Esc to cancel the edit.

For more information, see [Editing Digital Pin Data](#).

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Editing Pattern Data : Cutting, Copying, and Pasting Cells

Cutting, Copying, and Pasting Cells

Instead of entering data into vector cells, you can cut or copy cells and paste them elsewhere. This applies not only to pin data cells but also to patgen cells, that is, cells in the TSet, Command, and Label columns.

To cut or copy cell data:

- | | |
|----|--|
| 1. | Select a cell or range of cells. |
| 2. | Use the standard toolbar buttons, or Ctrl+X or Ctrl+C, or Edit>Cut or Edit>Copy to cut or copy the selected cells to the clipboard. The commands are also available on the right-click context menu. |

When you cut or delete a cell’s data, it is replaced with default data. For most digital pins, this is “0,” although some pin setups have a different default. For analog pins, the default is nop (blank). For TSets, commands, and labels, the default is blank. Note that TSet cells must have a value to be valid.

You cannot cut an entire pin column. However, you can drag a column to the position you want. To paste cell data in a new location:

- | | |
|----|--|
| 1. | Select the cells where you want to paste the data. |
| 2. | Use the standard toolbar button, Ctrl+V, or Edit>Paste. The command is also available on the right-click menu. |

For paste to function, the type of data on the clipboard must match the type of the cells selected to be overwritten. For example, you cannot paste Command data into a pin data cell, and you cannot paste the data from a pin group cell into an individual pin cell.

PatternTool fits the copied cells to the selected cells as well as it can. For example, if you copy three cells but select only two cells for pasting, it pastes into the two selected cells plus the next one.

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Editing Pattern Data : Editing TSet Cells

Editing TSet Cells

The TSet column displays the tset (time set) name or number for a vector. Before you can use a tset name, it must first be imported into the pattern file. PatternTool lets you view, edit, and add imported tset names using the Pattern Properties window.

A tset value must appear for every vector except serial scan vectors. If the vector is a serial scan vector, the Tset cell is disabled for data entry.

You can edit this column. For easier editing, double-click a cell to display a menu that contains all valid values, that is, all the imported tset names and any new names you have added. If you directly edit a cell and enter a name that has not been imported and is not on the menu, PatternTool adds it to the tset name table.

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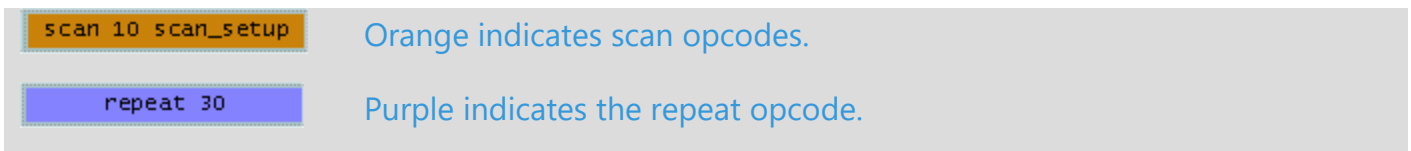


[Editing Pattern Data](#) : Editing Command Cells

[Editing Command Cells](#)

The Command column of PatternTool displays any PatGen microcode for a vector. PatGen microcode consists of an opcode or control bits or both. Note that Command cells display only PatGen microcode. Instrument microcode appears with the pin associated with the instrument, not in Command cells.

PatternTool uses color coding to make certain PatGen microcode stand out. In the PatGen opcode cells (labelled Command), these colors are used to highlight opcodes:



You can directly edit a Command cell.

For a quick edit, double-click in the Command cell. The cell becomes editable. This can be useful for changing a loop count or repeat count.

For more complex editing, you can use the command editor. To bring up the editor, right-click the Command cell and select Launch Editor. You can also select the cell and user Edit>Launch Editor.

The editor looks like this:

Repeat/Loop Branch Subroutine Misc Scan Counters

☒ repeat
 ☐ pipe_minus
 ☐ loop
 ☐ end_loop
 ☐ push_loop
 ☐ pop_loop
 ☐ clr_loop

Counters

☐ c0	☐ c4	☐ c8	☐ c12
☐ c1	☐ c5	☐ c9	☐ c13
☐ c2	☐ c6	☐ c10	☐ c14
☐ c3	☐ c7	☐ c11	☐ c15

Label:

Relative Address (+/-):

Number:

1000

Control Bits

☐ care
 ☐ mask
 ☐ stv
 ☐ refresh_en

Condition Set Flags Clr Flags Inv Flags

☐ branch_expr	☐ None	☐ fail	☐ pass
☐ iddq	☐ !iddq	☐ inst_cond	☐ !inst_cond
☐ cpuA_cond	☐ !cpuA_cond	☐ cpuB_cond	☐ !cpuB_cond
☐ cpuC_cond	☐ !cpuC_cond	☐ cpuD_cond	☐ !cpuD_cond

repeat 1000

OK Cancel

Command Editor

The editor displays all PatGen commands available for this vector. The editor automatically enforces rules governing each opcode. When you choose an opcode in the Opcode area, other areas of the editor are automatically enabled or disabled, depending on what syntax elements can be used with this opcode. For example, if an opcode does not take a number as an operand, the Number area is grayed out. If the opcode can be conditionalized, the check boxes for the various conditions are enabled.

Most opcodes can be used in both a VM and an SRM module, but some are SRM-only. The editor enables only those opcodes that are valid for the type of module that you are editing.

As you select or enter microcode elements, they are added to the read-only text box at the bottom of the editor. You cannot directly edit this text. You change it only by selecting, deselecting, entering, and deleting elements elsewhere on the dialog box. As soon as the current text is valid (meaning that the opcode has any required operand), the OK button is enabled. If it is invalid, the OK button is disabled.

The following are the areas of the editor, given in the recommended order for making selections:

Opcode tabs

Choose the tab for the opcode category, and then choose an opcode on the tab. Valid options for this opcode are then enabled on the editor, and invalid options are disabled.

To clear the current opcode, go to the Misc tab and select Nop. Nop clears the opcode plus any operand and conditions. It does not clear any

currently selected control bits because control bits can be used without an opcode. To add control bits without an opcode, choose Nop.

You can also delete the entire value from a Command cell in the main PatternTool display.

See [List of Opcodes and Control Bits](#) in the Pattern Language section.

Counters, Label,
Relative Address,
Number

If the selected opcode requires an operand, the appropriate operand area is enabled:

- For opcodes that set, reload, or use a counter, you must select a Counter. The reload opcode can act on several counters at once; otherwise, you can select only one counter.
- If the opcode takes a label operand, type in a label.
- If an opcode that takes a label is used in an SRM module, it may also take a relative address instead. If so, this field is enabled. You can enter a label or a relative address, but not both.
- If the opcode takes a numeric operand, type in a number. If you enter a number that is out of the valid range for this opcode, OK is disabled. Some opcodes can take either a counter or a number. In this case, you can select a counter or enter a number, but not both. On the Scan tab, the Number field becomes Scan Cycles.

Control Bits

You can add Control Bits to any vector, whether or not it has an opcode. (The exception is refresh_en, which can appear only once per vector group.) See [Control Bits](#) in the Pattern Language section.

Condition tab

If the selected opcode can be conditionalized, this tab is enabled. To conditionalize an opcode, select branch_expr on the Condition tab. The microcode appears as if (branch_expr) opcode.

To use a compound condition, first set up the branch_expr opcode (on the Branch tab). When you select branch_expr, you are setting up the compound condition; you are not conditionalizing the execution of the branch_expr itself.

A branch_expr can use any number of the conditions. You cannot select both complementary conditions (for example, cpuA and !cpuA, or fail

Set Flags, Clr Flags,
Inv Flags tabs

and pass). Compound conditions are joined by "and."
[See Conditional Statements](#) in the Pattern Language section.

[These tabs are with the](#) CPU Cond selection on the Misc tab. If you select CPU Cond, the three opcodes relating to CPU condition flags are enabled on the tab: set_cpu_cond, clr_cond_flags, and inv_cond_flags. As you select each opcode, the corresponding tab is enabled, letting you specify which flags are to be set, cleared, or inverted. Note that these opcodes can all appear on the same vector.

For more information, see [List of Opcodes and Control Bits](#) in the Pattern Language section.

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Editing Pattern Data : Editing Label Cells

Editing Label Cells

Cells in the Label column of PatternTool display any label for a vector. The format is type labelname: where type can be any valid label type. The colon is a required part of the label. If type is omitted, the label is a local label.

For a quick edit, double-click in the Label cell. The cell becomes editable.

For more complex editing, you can use the label editor. To bring up the editor, right-click the Label cell and select Launch Editor. You can also select the cell and user Edit>Launch Editor.

The editor consists of a grid containing any labels currently defined for the cell, plus a blank row for a new label. As you enter a new label or edit an existing one, the text box at the bottom of the editor displays the label. If the label is valid, OK is enabled. If the label is invalid, OK is disabled. A label can contain only letters, digits, and underscores and must begin with a letter.

Enter a valid name under Label. Under Type, select a type from the menu. Certain label types are valid only in an SRM pattern module and cannot be used in a VM module. If you are editing a VM module, the drop-down list contains only types valid in VM.

For more information, see Vector Labels in the Pattern Language section.

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Editing Pattern Data : Editing Digital Pin Data

Editing Digital Pin Data

For digital pins, the cells for each vector contain data for the pin. The data representation can be symbolic or numeric as determined by the radix of the pin, which you set using [Edit>Pin Structure](#). A dash (-) in this column indicates runtime repeat, that is, repeat the data for this column from the previously executed vector.

You edit a single pin data cell directly. If you enter an invalid value, PatternTool does not let you exit until you enter a correct value or press Esc to cancel the edit.

The following topics are available:

- [Pin Group \(Bus\) Data Edit Window](#)
- [Complementing and Swapping Values](#)
- [Replicating Values](#)
- [Shifting Values](#)

For more information, see [Symbolic and Numeric Pin Data](#) in the Pattern Language section.

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Editing Pattern Data : Editing Digital Pin Data : Pin Group (Bus) Data Edit Window

Pin Group (Bus) Data Edit Window

To edit data for a pin group or bus, as opposed to an individual pin in a group, double-click a cell in the pin group. This brings up a window for editing the group or bus data:

Format

☒ Symbolic

☐ All same

☐ Drive

☐ Receive

Radix

☒ Binary

☐ Octal

☐ Decimal

☐ Hexadecimal

Symbolic Data

01LHX

M

V-

Data:

01010101

OK

Cancel

Pin Group and Bus Editor

On the window, the current pin data for the cell is displayed in the Data box. When a value is valid, OK is enabled. PatternTool does not let you enter invalid characters. If you enter too many or too few characters, OK is disabled. You can use Delete to delete characters, but you cannot always directly enter characters. You must use the buttons, as follows.

The Data box displays data in the currently selected format. Changing the format enables different editing capabilities. Therefore, select the Format first. Format options are:

Symbolic

This format enables the Symbolic States buttons. When you click a button, that character is inserted at the position of the cursor in the Data box. If this results in too many characters, OK is disabled until you delete the necessary number of characters. You can delete characters in the Data box directly.

All same

If you use the mouse to select several characters in the Data box, clicking a Symbolic States button changes all selected characters to that value. See [Symbolic Vector Programming](#) in the Pattern Language section.

Drive, Receive

This format lets you set all pins to the same symbolic value. You can use a Symbolic States button to set the value. The Data box displays the single symbolic value, preceded by a period.

These formats use numeric values, prefixed by “.d” and “.r.” If you select one of these formats, the Radix buttons are enabled, letting you change the display radix. Select the radix, and then enter valid characters for that radix in the Data box.

See [Numeric Vector Programming](#) in the Pattern Language section.

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Editing Pattern Data : Editing Digital Pin Data : Complementing and Swapping Values

Complementing and Swapping Values

PatternTool includes commands to swap pin states in selected cells with certain other states. This allows you to:

- Swap values with their complements
- Swap drive data with receive data
- Swap specified pin data values for other pin data value

The Complement command is on the Edit menu and the right-click context menu. Select the range of cells whose values you want to complement, then select the Complement command. It changes each pin state to its complement: 0 to 1, 1 to 0, L to H, and H to L. It has no effect on the other datacodes.

The Swap command is on the Edit menu and the right-click menu. If desired, select the range of cells whose values you want to swap, then select the Swap command. This window comes up:

The image shows a software dialog box with three tabs: 'Find', 'Replace', and 'Swap'. The 'Replace' tab is selected. Inside the 'Replace' tab, there are three main sections:

- Options:** Contains three radio buttons: 'Complement', 'Swap 0/1 with L/H', and 'Replace States'. 'Replace States' is selected.
- Replace States:** Contains two text input fields labeled 'From:' and 'To:'. To the right of these fields are three buttons: 'Drive States', 'Receive States', and 'All States'.
- Search options:** Contains a section labeled 'Find in:' with three radio buttons: 'Selected range', 'Current module', and 'All files'. 'Selected range' is selected.

At the bottom of the dialog box, there are three buttons: 'Less', 'Swap', and 'Close'.

Complement and Swap Window

Window controls include:

Complement

Same as the Complement command.

Swap 0/1 with L/H

Swaps drive and receive data, replacing 0 with L, 1 with H, and vice versa.

Replace States

Enables controls on the right side of the dialog box. These controls let you specify:

- **From:** States to be changed. Type in one or more states.
- **To:** States to change to. Type in either one value or the same number of values as in From states.
- **Drive States:** Puts all drive states in the From field. You can edit these states.
- **Receive States:** Puts all receive states in the From field. You can edit these states.
- **All States:** Puts all states in the From field. You can edit these states.

You can replace each state with a specific single state or all states with a single state. For example, you can specify that 0 and L be replaced with 1 and H (respectively), or that H, M, and V all be replaced with X.

If you enter an equal number of From and To states, the nth From states value is replaced by the nth To states value. If you specify exactly one value for To, all the From states are replaced by the single To state value.

Search options

Specifies what the Swap is to act on: the currently selected range of cells, the current module, or all files currently displayed.

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Editing Pattern Data : Editing Digital Pin Data : Replicating Values

Replicating Values

The Replicate commands let you select one or more cells and then repeat the cells' values along the pins that contain the cells, starting immediately after the selected cells. You can replicate the values a specified number of times, to the end of the module, or to the end of the pin. Replication overwrites any existing data in the cells.

You can select a continuous range of cells covering multiple pins and vectors. You then select one of the three Replicate commands on the Edit menu or the right-click context menu:

Replicate...	Brings up a box that lets you specify how many times the cells are to be repeated. If the number of replications goes beyond the boundary of the module, it continues into the next module. The control sequence is Ctrl+R.
Replicate End of Module	Repeats the cells for the pin to the end of the current pattern module. The control sequence is Ctrl+M.
Replicate to End	Repeats the cells to the end of the pin. The control sequence is Ctrl+E.

Replicate can operate on any editable data in the pattern: pin data, TSets, commands, labels, and comments. You can use Replicate within a scan chain. You cannot replicate from non-scan vectors into a scan chain. You cannot undo a Replicate command.

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Editing Pattern Data : Editing Digital Pin Data : Shifting Values

Shifting Values

The Shift command lets you select one or more cells and move them up or down a specified number of vectors. The Shift command is on the Edit menu and the right-click context menu. The command brings up a window that lets you specify the following:

Number of vectors to shift	The number of minor vectors to shift.
Shift Cell(s) Up	Moves the selected cells one cell toward the first vector. Shifting cells up leaves one cell "empty." The empty cell is filled with the value from the following vector. When shifting up from the last vector, the shifted data remains in the last vector.
Shift Cell(s) Down	Moves them one cell toward the final vector. Shifting cells down leaves one cell "empty." The empty cell is filled with the value from the preceding vector. When shifting down from the first vector, the shifted data remains in the first vector.

Note these points about shift:

- | |
|--|
| <ul style="list-style-type: none">Shifted data overwrites any existing cell data. |
| <ul style="list-style-type: none">Data in the selected cells that are shifted does not change. |

- .Shift can operate on any editable data in the pattern: pin data, TSets, commands, labels, and comments.

Scan setup vector data is treated as any other vector, but data that is shifted into a scan vector is lost.

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Editing Pattern Data : Viewing and Editing Block Comments

Viewing and Editing Block Comments

The IG-XL pattern compiler supports block comments, which appear between vectors:

```
> - 0 1 00000100 LLLLLHLL ;
/* Multi-line block comment      */
/*      between vectors      */
> - 0 1 00000101 LLLLLHLH ;
```

PatternTool lets you view, add, and edit block comments.

 **Block comments** Use this button or the View>Show Block Comments command to toggle between showing and hiding block comments.

The button is enabled only if the pattern displayed contains block comments.

Note that a block comment is associated with a specific vector, and that the block comment appears before its vector. If the .atp file creates a block comment after the final vector, the comment is not associated with a vector and therefore is not displayed in PatternTool.

You can also use the Goto navigation control to go to the previous or next block comment. See Goto in the section on navigation.

To insert a block comment, select at least one cell in the vector that you want after the block comment, and then select Insert>Block Comment. The comment is inserted like this:

TSet	Command	Label	Vector	TD	Comment
TD1_ts1		start_label TD1...	0	00	
TD1_ts1	branch_expr = (!cpuA_cond)		1	00	
TD1_ts1			2	00	
TD1_ts1	clr_cond_flags (cpuA_cond)		3	00	
TD1_ts1		loop:	4	00	
TD1_ts1			5	11	
TD1_ts1			6	00	
Add comment here					
TD1_ts1	if (branch_expr) jump loop		7	11	
TD1_ts1			8	00	
TD1_ts1	halt		9	00	

Block Comment Between Vectors

You directly edit the comment. If the comment is longer than the width of the display, it wraps once you exit from the comment area.

To delete a block comment, select it and then select Edit>Delete>Vector or Comment or select Delete Vector from the right-click context menu.

If the ASCII source file for a pattern file contains comments and is compiled using the option Save Comments, the comments are stored in the compiled pattern file and can be viewed in PatternTool. If you use PatternTool to add comments, and if the save_comments option is enabled, these comments become part of the compiled pattern file and appear in the output of reverse compilation. If save_comments is no, comments you insert are not saved.

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Editing Pattern Data : Inserting, Copying, and Deleting Vectors

Inserting, Copying, and Deleting Vectors
The following topics are included:

- | | |
|---|---|
| • | Inserting Vectors |
| • | Cutting, Copying, and Pasting Vectors |
| • | Deleting Vectors |
| • | Moving Vectors to a New Subroutine |

Note that these commands do not apply when scan vectors are selected. See [Viewing and Editing Scan Chains](#).

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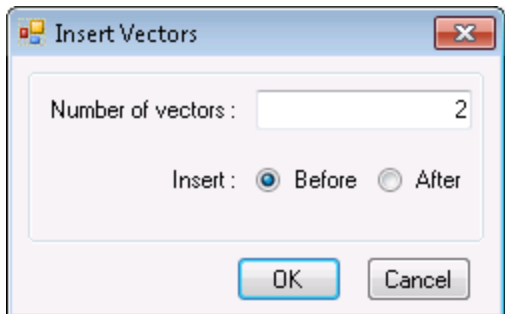
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Editing Pattern Data : Inserting, Copying, and Deleting Vectors : Inserting Vectors

Inserting Vectors

The command for inserting vectors is [Insert>Vectors](#). The command is enabled only when you have one or more entire vectors selected in the display area. The command brings up the Insert Vectors dialog box, which asks for the number of vectors to be inserted and whether to insert them before or after the currently selected vectors:



Insert Vectors Window

A right-click context menu item is also available for inserting vectors. Select a number of vectors equal to the number that you want inserted, then right-click and select [Insert Vectors](#).

Vectors are inserted in keeping with the opcode mode. For example, in dual mode, inserting one vector inserts two rows, which make up the major vector. In addition, the new vector is inserted before or after the entire current vector: it cannot be inserted between the two rows of a major vector. This shows a pattern before inserting, with vector 5 selected:

TSet	Vect	2 x	2 x	i n	i o	Comment
ts_dual_2x	0	00	XX	0	L	0
ts_dual_2x	1	01	XX	0	L	1
ts_dual_2x	2	10	XX	1	H	2
ts_dual_2x	3	11	XX	1	H	3
ts_dual_2x	4	00	LL	0	L	4
ts_dual_2x	5	01	LH	1	H	5
ts_dual_2x	6	10	HL	0	X	6
ts_dual_2x	7	11	HH	1	X	7

Before Inserting Vector

With vector 5 selected, if you request inserting one vector before the current vector, the two rows are inserted before vector 4; the rows are not inserted immediately before vector 5.

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Editing Pattern Data : Inserting, Copying, and Deleting Vectors : Cutting, Copying, and Pasting Vectors

Cutting, Copying, and Pasting Vectors

Instead of inserting new vectors, you can cut or copy existing vectors and paste them in a new location.

To cut or copy vectors:

1.

Select one or more entire vectors by clicking in the left-most cell in the display area.
2.

Use the standard toolbar buttons or Edit>Cut or Edit>Copy to cut or copy the selected vectors. The commands are also available on the right-click menu.

To paste vectors in a new location:

1.

Select the vector indicating the new location.
2.

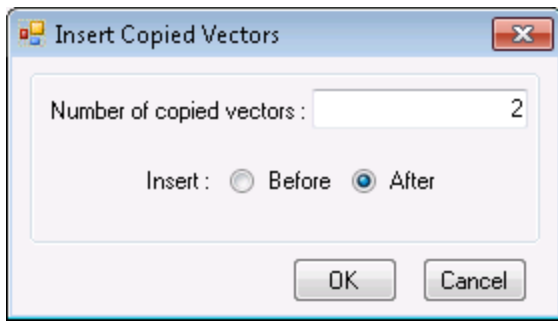
Do either of the following:

–

To replace the currently selected vectors with cut or copied vectors, use Edit>Paste (also available as the standard toolbar button and on the right-click menu). This command overwrites the selected vectors.

–

To insert cut or copied vectors after the currently selected vectors, use Insert>Copied/Cut Vectors, which is also available of the or right-click context menu. This command does not overwrite the selected vectors. It is enabled only when vectors have been cut or copied.



Insert Copied Vectors Window

You can specify how many times the vectors should be inserted and whether to insert them before or after the selected location.

The simplest paste is when the number of vectors you cut or copy equals the number of vectors you select to be overwritten. If they are not equal, PatternTool makes the most logical paste it can:

- If you copy one vector and select three, Paste pastes the one vector three times, overwriting the three selected vectors.
- If you copy three vectors and select the first, Paste overwrites the selected vector plus the two after it.
- If you copy two vectors and select four, Paste pastes the two vectors twice, overwriting the four selected vectors.

If you copy nonadjacent vectors, they are pasted or inserted as if they were a block of adjacent vectors.

			
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Editing Pattern Data : Inserting, Copying, and Deleting Vectors : Deleting Vectors

Deleting Vectors

To delete vectors, select the vector or vectors by clicking in the leftmost cell. Then use the Edit>Delete>Vector or Comment command. The Delete>Vector command is also on the right-click context menu.

You can delete nonadjacent vectors.

If you delete one vector from a vector group, all the remaining vectors move up one position. This may make the pattern invalid: for example, in a dual-mode pattern, it may move an opcode from a major to a minor vector, which is invalid. When you try to update the tester or save the file, you are told that the action cannot be performed because of the invalid opcode placement.

You cannot delete scan vectors using this command. If you include scan vectors in your selection, the delete command is not enabled. If you delete the vector containing the scan opcode, you delete the entire scan chain. For more information, see [Viewing and Editing Scan Chains](#).

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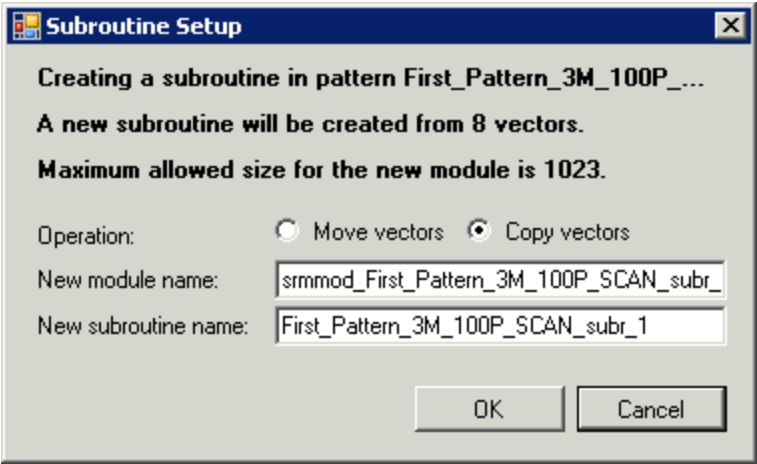
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Editing Pattern Data : Inserting, Copying, and Deleting Vectors : Moving Vectors to a New Subroutine

Moving Vectors to a New Subroutine

PatternTool provides a single command for creating loops or other code that requires use of SRM. Select the major vectors to be moved to a subroutine, and then execute the Edit>New Subroutine command. The following window appears:



Subroutine Setup Window

Indicate if you want the selected vectors moved or copied. Enter a nondefault name for the SRM module and for the subroutine. When you click OK, IG-XL does the following:

- Create a new SRM module, if one is not present in the pattern file.
- Check the SRM module for available space and notify if there is insufficient space.
- If space is available, copy the selected vectors to SRM, add a subr label with the subr name, and append a return opcode.

The command is enabled only if you have selected valid vectors. For example, if you include vectors from a vector group, you must include the entire group.

This command is a major edit that requires a recompile and reload of the pattern.

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Editing Pattern Data : Using Find and Replace

Using Find and Replace

PatternTool provides a dialog box with three tabs for finding a specified item, replacing an item with another, and going to a specified item. The commands are on the Edit menu, and each has a control-key equivalent:

- Edit>Find (Ctrl+F), see [Using Find](#)
- Edit>Replace (Ctrl+H), see [Using Replace](#)

The Find/Replace window has a third tab for Swap. See [Complementing and Swapping Values](#).

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Editing Pattern Data : Using Find and Replace : Using Find

Using Find

To find a specific item, use Edit>Find or Ctrl+F to bring up the Find tab. Find supports three kinds of searches, selected from the Find what: drop-down list:

Text	Finds the text entered in the box to the right of the Find what: control. Use this option to search for labels, opcodes, operands, timeset names, pin names, and comments.
Change in Data	Finds the cell whose data differs from the previous vector. The most common use of this option is to select a range of a pin's data and search it for a data value change.
Clipboard Data	Finds the cell or block of cells that match the data currently on the clipboard from a previous Copy.

These options are described in detail below.
In addition to these searches, the Find tab has this control:

Direction	Whether the search is Forward or Reverse from the currently selected cell, or whether the search is over Any cells regardless of the current selection.
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The More button displays more search options:

Find in top level items (e.g., Bus, Mux, etc.)	Whether to search only data that is visible when the display is not expanded.
Find whole words only	Whether the search string can match only whole words, or whether it can match part of words.
Match case	Whether the found text must match the case of the search text.
Find in	Specifies the search area:

- **Selected range:** Only the range of cells that you have selected. If this box is checked but no range is selected, Find finds nothing.
- **Current module:** All cells in the current module.
- **All files:** All currently open pattern files.

Use this option to restrict the area searched. For example, to search only labels, click the Label cell to select that row or column.

When Find what: is Text, you enter the string to be searched for in the text box. The string cannot use wildcards, although the string will act as a template and match parts of words, unless you have checked Find whole words only.

When Find what: is Change in Data, you do not enter a search string. This option highlights two adjacent cells where the data has changed from one vector to the next. For example, if a pin has "01" for one vector and "00" for the next vector, this search highlights the two cells. The search can cover all data in a pattern, including TSets, commands, labels, and pin data. You can restrict the area searched by checking Find in Selected Range.

When Find what: is Clipboard Data, you do not enter a search string. Instead, you select a cell or block of cells and use Edit>Copy (Ctrl+C) to copy it to the clipboard. Find then highlights the cells that match what's on the clipboard. The match must be exact. You can restrict the area searched by checking Find in Selected Range.

Note these points and suggestions:

- To find a TSet name, label, command, or comment, first select the entire column by clicking the header cell. For example, click the header Label to select all labels in a pattern. On the Find tab, set Find what: to Text. Enter the string to be found. Under the find options (revealed by the More button), check Find in Selected Range.

- [To find a block of cells, set](#) Find what: to Clipboard Data. Select a block of cells that has the pattern you want and use Edit>Copy (Ctrl+C) to copy it to the clipboard.
- [Find operates only on items in a pattern: pin data, labels, and so on. It does not operate on the headings that PatternTool uses to display the pattern. Specifically, it cannot find a pin or bus name, because these are display headings, not pattern items. To go to a specific pin or bus, use the navigate function set to Pin or Bus Name. See Goto.](#)
- [Find searches only exposed data. If individual pins in a bus are collapsed, it does not search the hidden data for the individual pins. In addition, Find searches only items contained in the current pin view setup. If the current setup omits a pin or other item, Find does not search the omitted items.](#)

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Editing Pattern Data : [Using Find and Replace](#) : Using Replace

Using Replace

To replace a specific item, use Edit>Replace or Ctrl+H to bring up the Replace tab. Replace lets you find one string of text and replace it with another. You specify the item to be replaced in the Find what: box. This is equivalent to the Find tab when Find what: is set to Text. The Find search options apply when used with Replace. (See [Using Find](#).) You specify the replacement text in the Replace with: box.

Use Find Next to find the next occurrence of a string, and Replace to replace it. You can also use Replace All to replace all occurrences of a string.

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Editing Pattern Data : Masking Transitions

Masking Transitions

You may want to mask output transitions, as well as I to O or O to I transitions, across certain pins and patterns. For example:

- Mask one cycle after the H to L transition:

HHHHLLLL->HHHHXLLL

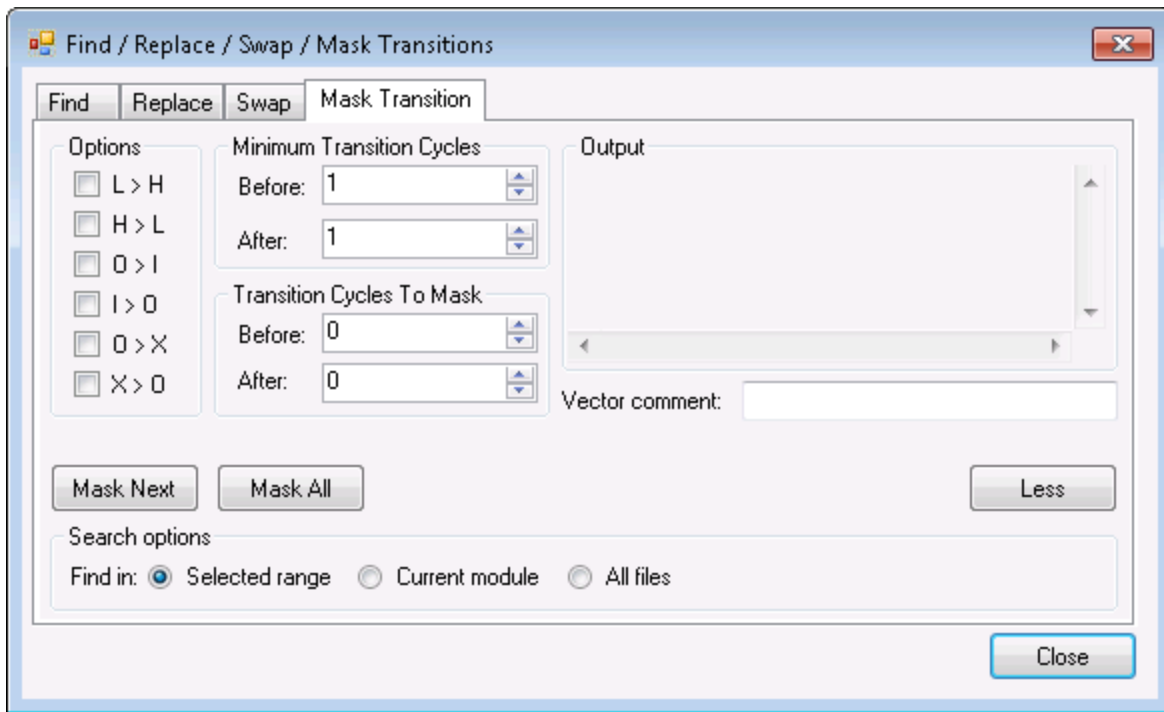
- Mask two cycles before the transition from L to H, plus three after it:

LLLLHHHH->LLXXXXXH

- At the transition from input to output, mask two cycles after the transition:

1111HHHH->1111XXHH

On the Edit menu, the Mask Transition command brings up this dialog box:



Mask Transitions Dialog

As you specify what you want masked, the Output pane displays what you have currently specified. Use the controls as follows:

1. Under Options, choose one or more types of transitions: L>H, 0>1, X>0, and so on.
2. Under Minimum Transition Cycles, specify how many consecutive cycles must match before and after the transition to determine if this is a transition to act on.

For example, if you select the L>H transition, with two cycles before and three after, the Output pane displays this: LLHHH > LLHHH.

3. Under Transition Cycles to Mask, specify how many cycles to mask before and after a matching transition.

For example, using the previous specification, if you specify to mask one cycle before and two after, the Output pane displays this: LLHHH > LXXXH.

The dialog box checks that the number of cycles to mask is smaller than the number of minimum transition cycles. Otherwise, the entire transition would be masked away.

4. Use Vector comment: to specify a comment that is to be appended to any masked transition.
5. If necessary, click the More button to display Search options. Use the radio buttons to specify the search range: the currently selected range, the entire current module, or all files currently

open.

6. When you have set up masking as you want it, execute it by clicking either of the following:

- **Mask Next:** Edits the next matching transition in the selected range.
- **Mask All:** Edits all matching transitions in the selected range.

You can abort the Mask Transitions operation while it is running. You can undo it when it has finished.

Mask Transition settings persist across sessions.

The following is a dialog box with maskings specified:

Options

☒ Low > High

☐ High > Low

☒ Output > Input

☐ Input > Output

☐ Output > X

☐ X > Output

Minimum Transition Cycles

Before: 4

After: 2

Transition Cycles To Mask

Before: 2

After: 1

Output

LLLLHH > LLXXXH
LLLL00 > LLXXX0, HHHH00 > HHXXX0, LLLL

Vector comment: Masked L>H, 0>I

Mask Next

Mask All

Less

Search options

Find in: ☐ Selected range ☐ Current module ☒ All files

Mask Transitions Specified

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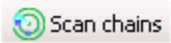
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Editing Pattern Data : Viewing and Editing Scan Chains

Viewing and Editing Scan Chains

PatternTool represents scan vectors in the main display area, but in a special way. Scan vectors in patterns can be compiled either as parallel vectors or as scan vectors to be loaded into the serial scan hardware option. PatternTool represents both the same way.

 Scan chains

You can turn display of scan chains on and off using this button or the View>Show Scan Chains command.

When the display is turned off, only the vector with the scan opcode is displayed:

ts0		12	L	L	L	L	L	000000	0L0L0L0L0L
ts0		13	L	L	L	L	L	000000	0L0L0L0L0L
	scan 5000	14							
ts0		15	L	L	L	L	L	000000	0L0L0L0L0L
ts0		16	L	L	L	L	L	000000	0L0L0L0L0L

Scan Vector with Scan Chains Off

If there are errors on a scan pin in the chain, its cell on this vector is colored.

When the display is turned on, the entire scan chain is shown:

ts0		12	L	L	L	L	L	000000	0L0L0L0L0L	0000000000
ts0		13	L	L	L	L	L	000000	0L0L0L0L0L	0000000000
	scan 5000	14								
		14:0	H	L	L	H	L			
		14:1	L	L	L	H	H			
		14:2	H	L	L	H	L			
		14:3	L	L	L	H	H			
		14:4	H	L	L	H	L			
		14:5	L	L	L	H	H			
		14:6	H	L	L	H	L			

Scan Vector with Scan Chains On

Each scan vector has the number of the scan opcode vector plus the number of the vector in the chain. The Tset, Command, and Label columns are disabled for the scan chain. Non-scan pins are

also disabled for data editing.

When scan chains are exposed, you can edit scan data as you do parallel pattern data.

To insert or delete scan vectors, edit the number on the scan opcode: double-click the cell to bring up the microcode editor. If you enter a number smaller than the current number, vectors are deleted from the end of the chain. If you enter a larger number, vectors are added to the end; the new vectors have default data. You can insert or delete scan vectors in the middle of the scan chain. To insert or delete individual vectors, you can use Shift with editing the scan opcode number. See Shifting Values.

To delete an entire scan chain, delete the vector that contains the scan opcode.

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Editing Pattern Data : Editing Instrument Microcode

Editing Instrument Microcode

If a pattern is set up to use extended microcode, the PatternTool pin data area contains a column for each instrument’s microcode. To assist in entering the microcode for an instrument, PatternTool provides a editor for each instrument. To bring up the editor, double-click the cell. For example, the editor for AWG6G looks like this:

AWG Src Microcodes

☒ Resync

☐ Stop

☐ StopE

☐ Enable_Inst_Cond

☐ Disable_Inst_Cond

☐ Start

☐ Start1

☐ StartE

☐ Start1E

☐ PSet

☐ Nop

Microcode: Resync

OK

Cancel

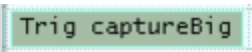
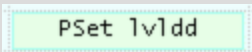
Instrument Microcode Editor for AWG6G

The editor assists in entering microcode by showing all available opcodes and by automatically enforcing rules and restrictions about their use. Note these points about using the microcode editors:

- As you select microcode for a cell, the currently selected microcode appears in the read-only Microcode text box. When the box displays the microcode you want, click OK.
- If opcodes can be used together, they are represented by check boxes, so that you can check as many as you want.

- If opcodes are mutually exclusive, they are represented by radio buttons, so that you can select only one of them.
- If an opcode requires an argument, when you select the opcode a text box appears after it. If the argument has a set of existing valid values, they appear on a drop-down list.
- The default opcode is "Nop," which is represented as a blank cell.

PatternTool uses colors to highlight the kind of instrument microcode contained in a cell:

	Dark green indicates opcodes that control the instrument (start, trigger, filter on or off).
	Blue indicates opcodes that collect, store, or transmit data.
	Pale green indicates the PSet opcode.

If you right-click a cell with the PSet opcode or with the Start or Trig signal opcode, the context menu includes Edit PSet/Signal, which brings up the PSet and Signals Editor with the PSet or signal displayed. See [PSet and Signals Editor](#) in the TDE section.

The information presented here applies to the editors for all instruments. For information on a specific instrument's dialog box, click [Related Topics](#).





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Opening and Managing Pattern Files

Opening and Managing Pattern Files

This section provides information on using PatternTool to open and manipulate pattern files. The following topics are available:

- [Opening, Saving, and Closing a File](#)
- [Stand-Alone PatternTool](#)
- [Importing and Exporting ASCII Files](#)

These sections focus on the commands available from the PatternTool File menu.

	
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Opening and Managing Pattern Files : Opening, Saving, and Closing a File

Opening, Saving, and Closing a File

PatternTool includes a set of file management commands, all available in the File menu. The commands are summarized here in the following sections:

- | | |
|---|---|
| • | Opening a File |
| • | Creating a New Pattern File |
| • | Saving and Closing a File |

For more information, see [File Menu Commands](#).



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Opening and Managing Pattern Files : Opening, Saving, and Closing a File : Opening a File

Opening a File

For opening a file, the options are:

Open	 Ctrl+O Opens a pattern file. A browser appears to let you navigate to the file. You can search for regular pattern files (.pat) or compressed pattern files (.pat.gz).
Open Last Burst Pattern	 Open Last Burst Opens the most recently burst pattern file, module, or set. The burst may have come from the test program, from the Immediate window, or from another tool, such as Pattern Control Display. This command displays whatever was most recently burst.
Recent Patterns	Displays a pull-right list of the most recently opened pattern files.
Reload from Saved File	Ctrl+Shift+O Reloads the currently displayed file from disk. This is the result of the most recent Save that you have issued or the original file that you opened if you have not used Save. You can use this command to undo all changes that you have made since the last Save. The command is also a shortcut to reopen the file from disk and reload it to the tester to synchronize vector memory.

You can have only one file open at a time in a single PatternTool, so opening a file requires closing any currently open file. If you edit a pattern file and then try to open another file, you are prompted to save your changes before closing the first file and opening the second.

You can view multiple pattern files simultaneously by bringing up multiple instances of PatternTool. In stand-alone mode, Open Last Burst Pattern and Reload from Saved File are disabled. See Stand-Alone PatternTool.

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Opening and Managing Pattern Files : [Opening, Saving, and Closing a File](#) : Creating a New Pattern File

Creating a New Pattern File

[PatternTool](#) lets you create a new pattern file.

 Use this button or the command [File>New](#). The New Pattern File window appears and prompts for this information:

Module name	Name of the pattern module.
Module type	Either VM or SRM.
Timing mode	Depends on the instrument type selected by the next control.
Instrument type	The drop-down lists the available digital instruments. You cannot create a new pattern for a serial instrument.
Vector count	Number of vectors in the module. The default is 64.
Insert pin	The drop-down lists the pins in the pin map. Select pins that you want inserted into the new pattern.

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Opening and Managing Pattern Files : Opening, Saving, and Closing a File : Saving and Closing a File

Saving and Closing a File
For saving a file, the options are:

Save	 Ctrl+S Saves the current pattern file to its current name and folder.
Save As	Saves the current pattern file to a new file. The new file becomes the current pattern file. The browser has a check box Generate unique names. If you check this box, the compiler appends an extension to all global symbols. This allows the saved file to be loaded into the tester without causing a symbol naming conflict with the original pattern. The algorithm takes the existing module names and labels and appends a string based on the current data and time.

When you save a pattern file that you have edited, PatternTool recompiles the file. Part of compilation is to check for errors: for example, if a pattern’s opcode mode is incompatible with the way opcodes are placed in vector groups. If a compilation error occurs, you are given the following options:

- Cancel the save, so that you can fix the problem before saving.
- Save the pattern anyway. If you select this option, the Save As dialog box is presented. Patterns that are saved with compilation errors are not loadable or executable. You should

assign names that reflect their incomplete state. You can use this option if you need to leave the tester with a pattern in an incomplete state.

If you create a new filename or pathname for a pattern file, note that the characters "+" (plus sign) and "," (comma) are invalid both in the pattern file filename and in the pathname to the pattern file. In addition, if a pattern file contains a single module that uses the default module name, the pattern file name must use only alphanumerics and the underscore and begin with an alphabetic character. See [Pattern File Filename and Pathname](#) in the [Pattern Language](#) section.

For closing a file, the options are:

Close File	Closes the current pattern without exiting from PatternTool.
Exit	Closes the current pattern file and exits from PatternTool.

When you close a pattern file, if there are unsaved changes, you are prompted to save them. In this case, the comment about validation on saving files applies to closing files as well.

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[Opening and Managing Pattern Files](#) : Stand-Alone PatternTool

Stand-Alone PatternTool

You can also bring up PatternTool in stand-alone mode, not connected to a test program that is executing patterns. You can use the [Start](#) menu:

[Start](#)>[All Program](#)>[Teradyne IG-XL Vxx.xx.xx](#)>[Tools](#)>[Pattern Tool](#)

You can double-click a .pat file in Windows Explorer, which opens the file in stand-alone mode.

When you have a pattern open in connected mode, you can open the same pattern in stand-alone mode using the PatternTool command [Tools](#)>[PatternTool \(StandAlone\)](#).

In stand-alone mode, you use PatternTool for viewing and editing pattern files. You cannot burst patterns because no test program is available to burst the pattern. Therefore, any features related to bursting patterns are disabled:

- [All commands on the Debug](#) menu and the corresponding toolbar buttons are disabled. See [Debug Menu Commands](#) and [Bursting Patterns](#).
- [The File](#)>[Open Last Burst](#) command and corresponding button are disabled. See [Opening, Saving, and Closing a File](#).
- [The File](#)>[Reload from Saved File](#) command is disabled. The command reloads the pattern from disk into PatternTool, but also reloads the pattern to the tester. Because there is no connection to the tester at this point, the command is disabled. See [Undoing and Redoing Edits](#).

- [The ability to open digital tools from the](#) Tools menu is not supported. It is not meaningful to open any of these tools when the pattern is not connected to an executing test program. See [Tools Menu Commands](#).
- [In the Pin Explorer, the only built-in setup is](#) All Pins. Because the pattern is not burst, there are no Passing Pins or Failing Pins. See [Using View Setups](#) in the section on the Pin Explorer.
- [Mask Transition](#) on the Edit menu is a hardware feature and is therefore not available with standalone PatternTool. See [Masking Transitions](#) in the section on debugging with PatternTool.

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Opening and Managing Pattern Files : Importing and Exporting ASCII Files

Importing and Exporting ASCII Files

PatternTool is used primarily for opening, editing, and saving binary files, that is, pattern files that are already compiled. You can also import an ASCII source file (.atp) into binary format in PatternTool, as well as write out a binary pattern file into ASCII format.

To import a file, use File>Import From Ascii. This command brings up a dialog box similar to the one for the Pattern Compiler. For information on filling in the fields, see [Pattern Compiler Window](#) in the Pattern Compiler section.

Compiler options are presented in the same kind of grid with drop-down lists of valid values. For more information, see [Compiler Control Statements](#) in the Pattern Language section.

Once you fill in the required values and select the desired options, you can click the button for the action you want performed:

Compile	Compiles the pattern file without opening it in PatternTool.
Open Pattern	Opens the most recently compiled pattern file.
Import	Compiles the pattern file and opens it in PatternTool.
Close	Closes the dialog box.

To export the currently displayed binary pattern file to an ASCII .atp file, use File>Export To Ascii. A window comes up displaying the options found in the Pattern Reverse Compiler. For more information, see [Pattern Reverse Compiler Overview](#) in the Pattern Compiler section.

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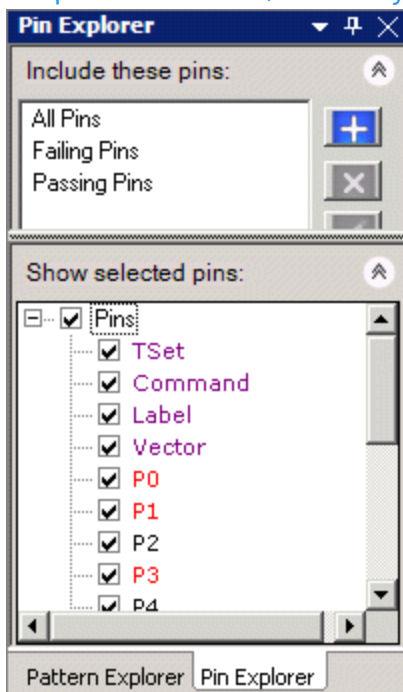
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Using the Explorer and Properties Windows

Using the Explorer and Properties Windows

This section provides information on using the Pattern and Pin Explorers. By default, these features appear as two tabs in a single panel on the left side of the tool. The section also covers the Properties window, which by default is under the Pattern and Pin Explorer panel:



Pattern and Pin Explorers

If they are not visible, you can use the View>Pin Explorer, View>Pattern Explorer, and View>Properties commands to make them visible.

You can use the Pattern and Pin Explorers to select what is to be displayed in PatternTool. You can use the Pattern Explorer to add or delete pattern modules, and the Pin Explorer to add, delete, or mask pins. The Properties window displays the properties of the currently selected pattern file, pattern module, or pin. Some properties are editable, which lets you change values and rebursting for debugging.

The following topics are covered:

- [Pattern Explorer](#)

- [Pin Explorer](#)

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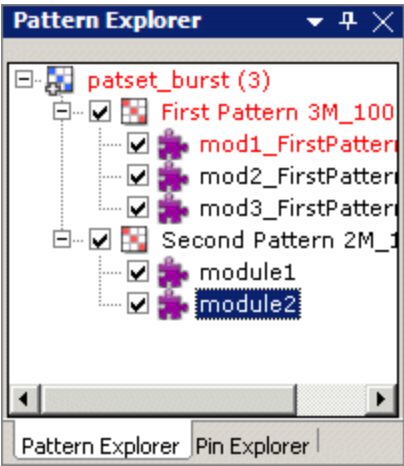
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Using the Explorer and Properties Windows : Pattern Explorer

Pattern Explorer

The Pattern Explorer shows the patterns and modules that are currently displayed in PatternTool:



Pattern Explorer

You can use the Pattern Explorer to do the following:

- View the names and structure of the currently displayed pattern sets, patterns, and modules, with failing modules highlighted
- Navigate to a selected module in a pattern
- Hide modules or patterns from the display and therefore from execution
- Add new modules to a pattern and rename and delete existing modules

- [Select a pattern or module whose properties are to be displayed and modified in the Properties window](#)

The following topics are available:

- [Contents of the Pattern Explorer](#)
- [Inserting, Renaming, and Deleting Modules](#)
- [Viewing and Changing Pattern Properties](#)

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Using the Explorer and Properties Windows : [Pattern Explorer](#) : Contents of the Pattern Explorer

Contents of the Pattern Explorer

The Pattern Explorer can be considered a top-level summary of what is in the display. The content of Pattern Explorer depends on how you opened the currently displayed pattern:

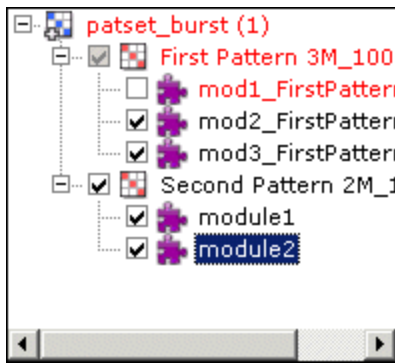
- If you used [File>Open](#), the Pattern Explorer shows the file and all of its modules.
- If you used [File>Open Last Burst Pattern](#), the Pattern Explorer shows the patterns and modules defined by the pattern set, plus any init patterns, referenced subroutines, and nondefault stop subroutines.

Note these points about pattern sets:

- A pattern set that uses one module of a file with multiple modules shows only that module. Other modules in the file are not indicated. If you open a pattern file itself, all of its modules are shown.
- A pattern set that contains repeated modules or files shows those items only once.

The Pattern Explorer shows what is potentially executable. It does not control what is burst or in what order. Different modules may be executed in a different order when the [Burst](#) command is executed. Modules are presented in terms of their order within the file, which is not necessarily execution order.

You can hide a pattern file or module by unchecking the box next to its name. For example, [mod1_FirstPattern](#) is hidden in the following figure.



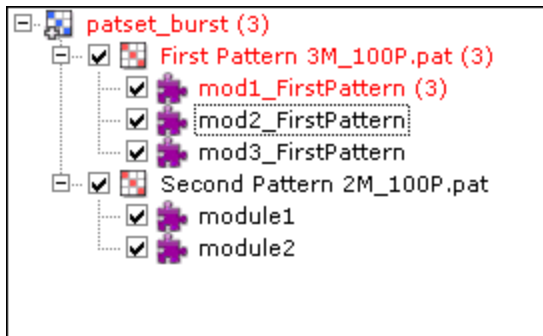
Pattern Explorer with Module mod1_FirstPattern Hidden

If you uncheck a module but leave others in the pattern file checked, the pattern file checkbox is grayed out.

Hiding a pattern file or module in the Pattern Explorer removes it from view in the display.

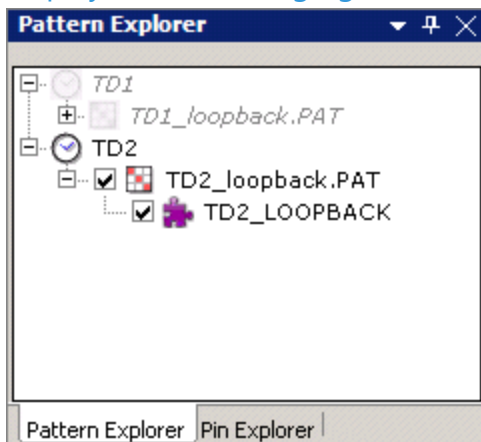
You can use the Pattern Explorer to display a specific module. Double-click the module name. The current view automatically scrolls to make that module visible. If you double-click a pattern file name, the current view displays the first module of that file.

Pattern sets, files, and modules that contain failures are displayed in red. The fail count for the item is given in parentheses after the item name. For example, mod1_FrstPattern in the following figure has three failures.



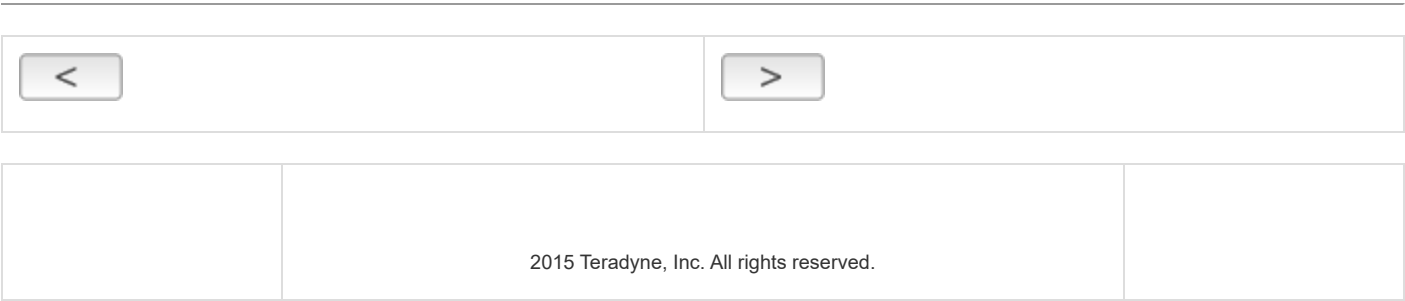
Pattern Explorer with Module mod1_FrstPattern Showing Failures

If the burst included multiple time domains, the time domains are the root of the Pattern Explorer display. The following figure shows two time domains, TD1 and TD2.



Time Domains in Pattern Explorer

PatternTool displays only one time domain at a time. Other time domain names are shown in gray. In the figure, TD2 is displayed, while TD1 is grayed out. To display a different time domain, double-click it. You can also click a time domain and select either of these commands from the context menu: Open <time domain> or Open <time domain> in new PT window.



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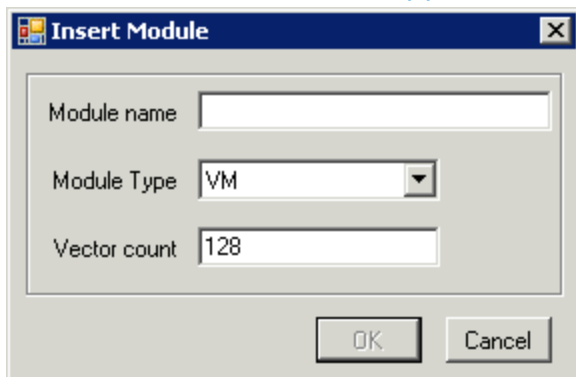
Using the Explorer and Properties Windows : Pattern Explorer : Inserting, Renaming, and Deleting Modules

Inserting, Renaming, and Deleting Modules

You can insert a module using Pattern Explorer as follows:

1. Select a pattern or module by clicking it in the Pattern Explorer.
2. Use the Insert>Module command, which is also on the right-click context menu.

The Insert Module window appears:



Insert Module Window

3. Enter the name of the new module, its initial size, and its type.

The module is appended to the chosen pattern, or inserted ahead of the chosen module.

To rename a module:

1. Select the module in the Explorer.

2. [Right-click and select](#) Rename Module from the context menu.

You can also rename a module in the Properties window. You cannot rename an implicitly loaded module, that is, a module that is loaded as part of a pattern set or pattern file that has been burst. PatternTool cannot update the tester with an implicitly loaded module that has been renamed, so it does not permit renaming. It returns an error if you try to rename such a module; you must use Esc to exit from the edit.

To delete a module:

1. [Select the module in the Pattern Explorer.](#)
2. [Use](#) Edit>Delete>Module or right-click and select Delete Module from the context menu.

You cannot delete the last module from a pattern: a pattern file must contain at least one module.

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Using the Explorer and Properties Windows : [Pattern Explorer](#) : Viewing and Changing Pattern Properties

Viewing and Changing Pattern Properties

Patterns and their modules have properties that you can modify. To modify pattern properties:

1. [Click a pattern or module in the Pattern Explorer.](#)
2. [Go the Properties window to view the properties for modules and patterns that are available for modification.](#)

For a module, the following properties are displayed:

Name	The name of the module, which is the only module property that you can modify, subject to a restriction. See Inserting, Renaming, and Deleting Modules .
IsLoaded	Whether the module is loaded: True or False.
ModuleType	The type of module: VM or SRM.
VectorCount	The number of vectors in the module.

For a pattern, properties are divided into two categories: [Compiler Options](#) and [General](#). You can use toolbar buttons on the Properties window to switch between categorized and alphabetical listing.

[Compiler Options](#) are those described as [Compiler Control Statements](#) in the [Pattern Language Reference](#) or as [Pattern Compiler Options](#) in the [Pattern Compiler](#) section. You can modify all

options except `digital_inst`. Options with a defined set of values have those values on a drop-down list. For other options, you can enter a value.

A compiler option can be specified either in the `.atp` file as a compiler control statement or at compile time. Under the General properties is File Statement, which indicates whether the option is specified in the file. See the description below. In most cases, the pattern source does not include compiler statements, so a default is displayed.

Changing compiler options has no immediate effect on the pattern file that is currently open. The options may affect a pattern file in the following cases:

- Some compiler options, such as `pinmap_workbook` and `pinmap_sheet`, have an effect if you reverse-compile and then recompile the pattern file.
- `opcode_mode` affects opcode placement in the vector group. If you change `opcode_mode`, you may invalidate your opcode placement. If this happens, a compilation error occurs when you try to save the file.
- If you add comments to a pattern file, make sure that the `save_comments` option is enabled. See [Saving Comments in a Compiled File](#) in the pattern language reference.

The General properties for a pattern file are:

Name	Read-only. Use Save As to change a pattern's name.
Path	Read only. Use Save As to change a pattern's path.
Imports	Shown as Collection.... Click to invoke an editor. The editor lists the imported tsets and labels, showing the type and name of each. To add a new import, use the blank row at the bottom of the editor. To delete one, delete the name in the Symbol column.
File Statement	Shown as Collection.... Click to invoke an editor. The editor lists all compiler options, followed by yes (the option was specified as a compiler control statement in the <code>.atp</code> source file) or no (the option was specified at compilation time, using the Pattern Compiler GUI or command line). Changing the value affects reverse compilation of a file. If the value is yes, reverse compilation recreates the statement in the output <code>.atp</code> file; if no, it does not create the statement.

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Using the Explorer and Properties Windows : Pin Explorer

Pin Explorer

The Pin Explorer lets you show or hide individual pins and groups and create or apply a “view setup” to the PatternTool display. Three setups are built into the tool: All pins, Failing pins, and Passing pins. Selecting a view setup causes the pin list to change to display the indicated pin types. You can use the Pin Explorer to do the following:

- View the individual pins, pin groups, and pin lists
- Create a view setup, a subset of pins that you want to view
- Navigate to a selected pin in the pattern
- Hide pins from the display and therefore from execution
- Add new pin to a pattern and rename, delete, and mask existing modules
- Select a pin, group, or list whose properties are to be displayed and modified in the Properties window

The following topics are available:

- Contents of the Pin Explorer
- Inserting, Deleting, and Modifying Pin Items

- [Using View Setups](#)

- [Viewing and Changing Pin Properties](#)

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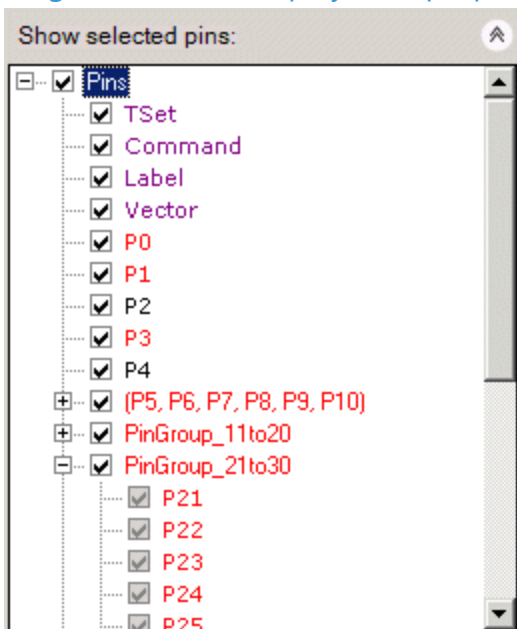
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Using the Explorer and Properties Windows : [Pin Explorer](#) : Contents of the Pin Explorer

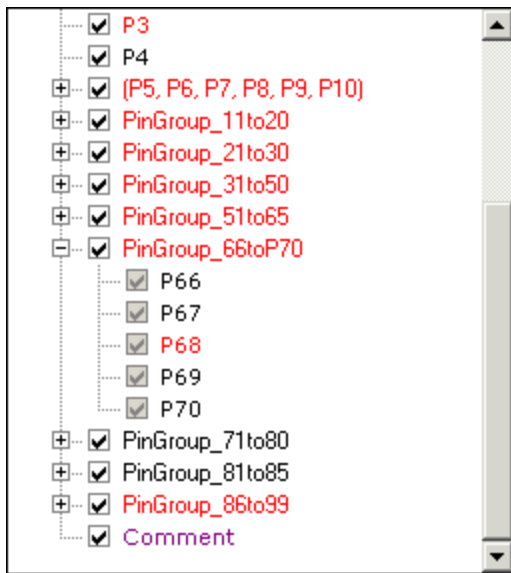
Contents of the Pin Explorer

The Pin Explorer shows the pins, groups, and lists in a pattern, including items for PatGen data. Patgen items are displayed in purple.



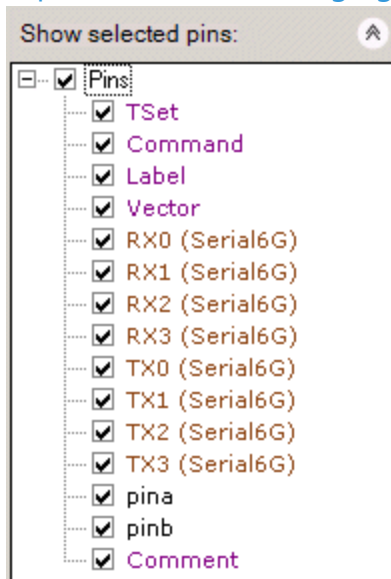
Pin Explorer

Failing pins are indicated in red. If any pin in a group or list fails, the name of the group or list is also in red:



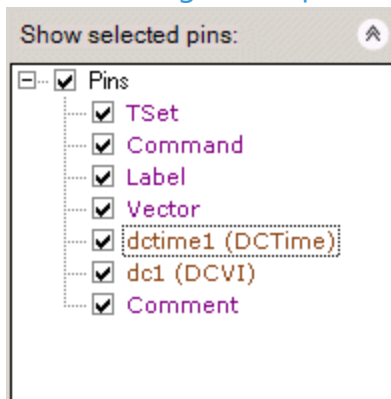
Pin Explorer Showing Failing Pins

Non-HSD pins show their instrument in parentheses after the pin name. For example, the Pin Explorer in the following figure includes SB6G pins:



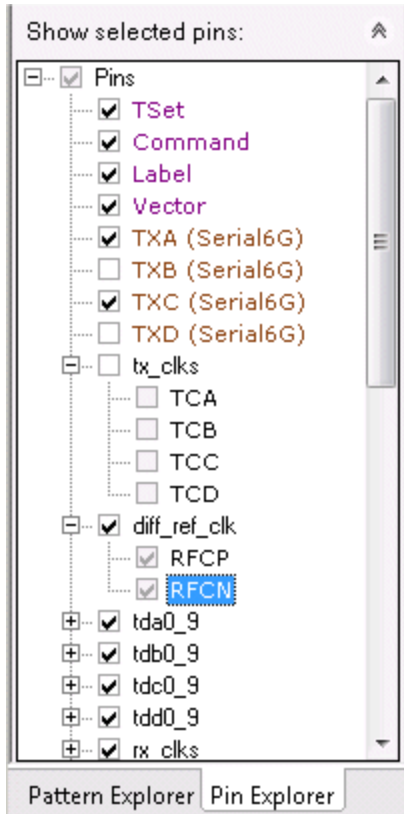
Pin Explorer Showing SB6G Pins

The following shows pins for DCVI and DCTime.



Pin Explorer Showing DC Pins

You can use the Pin Explorer to display a specific pin. Double-click the pin name. The current view automatically scrolls to make that pin visible. If the pin is part of a group or list, it is expanded so that the pin is visible. If you double-click a group or list name, the view scrolls to the group or list. You can hide a pin, group, or list by unchecking the box next to its name:

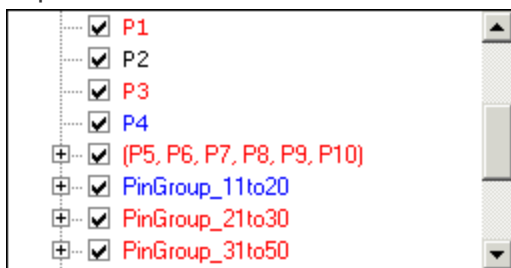


Pin Explorer Showing Unchecked Pins and Group

You cannot uncheck pins that are part of a pin group or list. The checkboxes for pins in the group or list are therefore disabled. You can only hide the entire group or list.

Hiding a pin, group, or list removes it from view in the display. It also removes it from execution by the Burst command.

Instead of hiding a pin, group, or list, you can mask it to remove it from the burst. See [Masking Pins and Vectors](#) in the section on bursting patterns. A masked item is displayed in blue in the Pin Explorer:



Pin Explorer Showing Masked Pins

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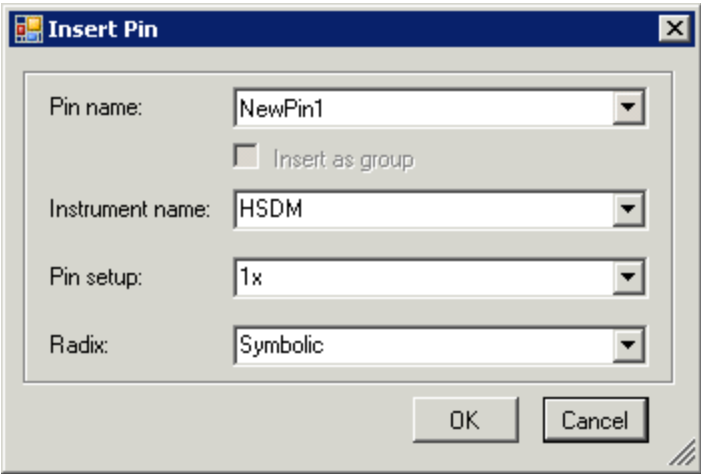
Using the Explorer and Properties Windows : Pin Explorer : Inserting, Deleting, and Modifying Pin Items

Inserting, Deleting, and Modifying Pin Items

You can insert or delete pins, groups, and lists. Adding or removing pins affects the binary pattern file that is open. The display grid is updated with any changes of this sort that you make. You can add a pin to all patterns that are open:

1. [Select a pin in the Pin Explorer.](#)
2. [Execute the Insert>Pin command](#), which is also available on the right-click context menu.

The Insert Pin window appears.



Insert Pin Window

3. [Enter a name. The drop-down list displays all pins and groups not currently used in the pattern.](#)

4.

Select the items you want by checking the box next to their names. If you select more than one pin, the checkbox Insert as group is enabled, letting you insert pins as a parenthesized list.

5.

Use the other controls to specify the instrument, setup, and radix. The new item is inserted above the pin you had selected in the Pin Explorer.

To delete a pin, group, or list:

1.

Select the pin, group, or list in the Pin Explorer.

2.

Execute Edit>Delete>Pin. The command is also available on the right-click context menu.

You cannot modify pin groups that are defined on a Pin Map sheet: you cannot insert, rename, or delete pins in the group. The right-click context menu lets you move a pin up or down in the group.

You cannot delete a bus that contains a scan pin. Trying to do this generates an error.

You can modify pins in a parenthesized pin list. Insert, delete, rename, and move up and down are all available for a pin list.

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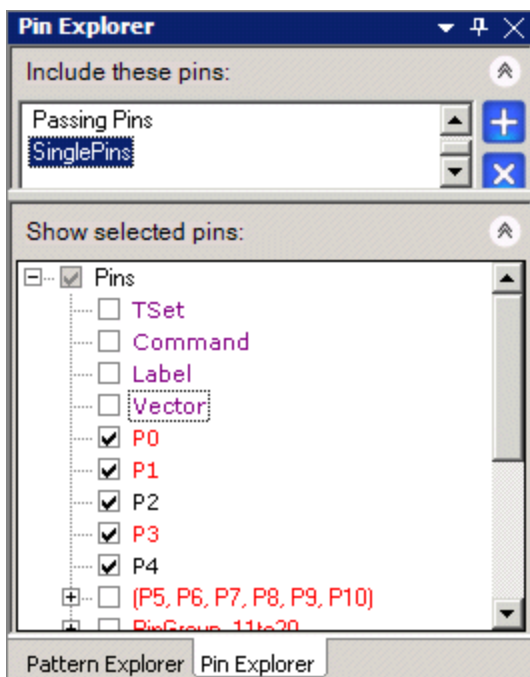


Using the Explorer and Properties Windows : Pin Explorer : Using View Setups

Using View Setups

You can use the Pin Explorer to create view setups. A view setup lets you select pins and groups that you want included in the rows of the display and save the selection as a named setup. You can later redisplay those rows by clicking the name of the setup.

For example, the view setup Single Pins in the figure displays only pins that are not part of a group or list, and omits PatGen data.



SinglePins View Setup

The built-in setups are All Pins, Failing Pins, and Passing Pins. You cannot modify or delete these setups. In stand-alone mode, only All Pins is available. See [Stand-Alone PatternTool](#).



To create a view setup, select the items that you want included by checking their boxes in the Pin Explorer. Then click the Add button. A new string is added to the View Setup panel with the default name "setupN." The name is in edit mode, so you can change it immediately.

Once you add a setup, you can double-click the name to display the selected rows.

 To delete a setup, select it in the View Setup panel and click the Delete button. To delete all user-created setups, right-click in the white area of the View Setup panel (not on a setup name) and select Delete All.

 To modify a setup, select it in the View Setup panel. Make the changes to the check boxes. Then click the Update button.

To rename a setup, select it in the View Setup panel, right-click, and select Rename. Note that Apply, Delete, and Update are also on the right-click menu.

		
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Using the Explorer and Properties Windows : Pin Explorer : Viewing and Changing Pin Properties

Viewing and Changing Pin Properties

Pins have properties that you can modify. To modify pin properties:

1.

Select the pin you want to modify in the Pin Explorer or in the main display.
2.

Go the Properties window to see the properties that are available for modification.

If you change a pin property, the change is applied immediately. Invalid values are immediately trapped and indicated by filling the cell with the red error color.

Pin properties are divided into several categories. There is a General category, and then other categories depending on the instrument. You can use toolbar buttons on the Properties window to switch between categorized and alphabetical listing.

The list of pin properties is instrument-specific. For nondigital instruments, the list can include:

General	Instrument name and pin name, both modifiable.
Instrument	Channel type (such as Input or Output), modifiable
Tester Channel	Channel to which the pin is connected, in both Signal and Pogo format, read-only.

Note that properties are displayed for nondigital pins only if they are selected in the Pin Explorer, not in the main display.

For digital pins, the list of available pin properties is context sensitive. If you select a pin in the Pin Explorer, timing properties for the pin are not available because no tset information is available. If

you select a cell of that pin in the main display area, timing information for the appropriate tset can be displayed and modified.

For digital pins, the list of properties includes:

General	Instrument name and pin name, both modifiable; IsMasked, modifiable; Fail count, read-only.
Levels	The pin levels, all modifiable.
Pattern	The representation of the pin in the pattern: pin setup and radix, both modifiable; whether this is a scan pin, read-only.
Tester Channel	The channel to which the pin is connect, in both Signal and Pogo format, read-only.
Timing/Format	Only if the pin is selected in the main display, not the Pin Explorer: some read-only, other (init and start states, format, edges, clock period, period) modifiable.

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PatternTool Toolbar and Command Reference

PatternTool Toolbar and Command Reference

This section describes PatternTool commands that are available on its menus and toolbars. PatternTool commands are arranged by the menu on which they appear:

- [Toolbars and Buttons](#)
- [File Menu Commands](#)
- [Edit Menu Commands](#)
- [View Menu Commands](#)
- [Insert Menu Commands](#)
- [Debug Menu Commands](#)
- [Tools Menu Commands](#)
- [Window Menu Commands](#)

Note that this description applies to PatternTool when used in the context of an executing test program. If you bring up PatternTool in stand-alone mode, many of the commands and button are disabled. See [Stand-Alone PatternTool](#).

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PatternTool Toolbar and Command Reference : Toolbars and Buttons

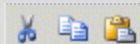
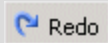
Toolbars and Buttons

Many PatternTool commands have equivalent buttons on the PatternTool toolbars. The following are the buttons on the toolbars, from left to right, with the equivalent command:

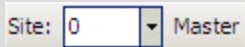
The first toolbar contains buttons mainly for controlling and editing pattern files:



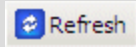
First PatternTool Toolbar

	File>New or Ctrl+N Opens a new pattern file. The New Pattern File window appears to let you specify the file. See Creating a New Pattern File.
	File>Open or Ctrl+O Opens an existing pattern file. See Opening a File.
	File>Save or Ctrl+S Saves the current pattern file to its current name and folder. See Saving and Closing a File.
	Edit>Cut/Copy/Paste or Ctrl+X, Ctrl+C, Ctrl+P Cuts or copies the selected data, then pastes it. See Cutting, Copying, and Pasting Vectors and Cutting, Copying, and Pasting Cells.
	Edit>Undo or Ctrl+Z Undoes the previous edit. The command indicates what the next Undo would affect: single or multiple cell edits, a shift, or a paste. See Undoing and Redoing Edits.
	Edit>Redo or Ctrl+Y

Redoes the previous undo. The command indicates what the next Redo would affect: single or multiple cell edits, a shift, or a paste. See [Undoing and Redoing Edits](#).



Selects the site whose data is to be displayed. If Master is clicked, uses the master site as set in DataTool. See [Selecting a Site](#).



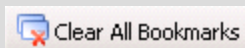
View>Refresh or F12

Refreshes the display after the pattern has been reburst from outside PatternTool. See [Display Basics](#).



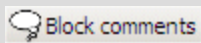
Edit>Toggle Bookmark or Ctrl+F2

Adds a bookmark to the currently selected vector, if it does not have one. Removes the bookmark if it has one. See [Bookmarks](#).



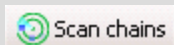
Edit>Clear All Bookmarks or Shift+F2

Removes all bookmarks. See [Bookmarks](#).



View>Show block comments

Exposes or hides block comments in the main display area. See [Viewing and Editing Block Comments](#).



View>Show scan chains

Exposes or hides scan chains in the main display area. See [Viewing and Editing Scan Chains](#).

The second toolbar has buttons mainly for bursting but also to control aspects of the display:



Second PatternTool Toolbar



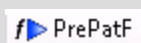
Debug>Update Tester or Ctrl+U

Writes the current pattern file, including any edits, to tester memory. See [Updating the Tester](#).



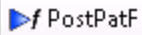
File>Open Last Burst

Opens the most recently executed pattern module, file, or set. The burst may have come from the test program, the Immediate window, or another tool. See [Opening a File](#)



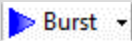
Debug>PrePatF

Toggles whether to execute the test's PrePatF interpose function before each burst of the pattern. See [Burst Options](#).



Debug>PostPatF

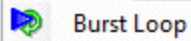
Toggles whether to execute the test's PostPatF interpose function before each burst of the pattern. See Burst Options.



Debug>Burst or Ctrl+B

Bursts the currently displayed pattern once. See Burst Options.

Burst is the default Burst option. You can access the other options by clicking the triangle.



Debug>Burst Loop

Bursts the currently displayed pattern in a loop. See Burst Options.



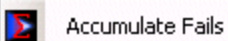
Debug>Burst Until Pass

Bursts the currently displayed pattern in a loop, until the pattern passes. See Burst Options.



Debug>Burst Until Fail or Ctrl+Shift+B

Bursts the currently displayed pattern in a loop, until the pattern fails. See Burst Options.



Debug>Accumulate Fails

Repeatedly bursts the currently displayed pattern to accumulate the fails from the entire pattern. See Accumulate Fails.



Debug>Halt or Ctrl+T

Stops execution of a pattern. See Burst Options.

Capture: Fails

Lets you specify what to capture in HRAM: Fails, FirstModFail, or STV. See Setting What To Capture.



Lets you navigate to the first, previous, next, or last instance of the selected item. See Navigation.



Debug>Learn or Ctrl+L

Starts Learn mode, which lets you ignore pattern errors and update the pattern directly using HRAM. See Using Learn Mode.



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PatternTool Toolbar and Command Reference : File Menu Commands

File Menu Commands

New	 Ctrl+N Opens a new pattern file. The New Pattern File window appears to let you specify the file. See Creating a New Pattern File .
Open	 Ctrl+O Opens an existing pattern file. See Opening a File .
Recent Patterns	Displays a pull-right menu of recently opened files.
Reload from Saved File	Ctrl+Shift+O Loads the pattern from disk into PatternTool, deleting any changes made since the last Save or since the file was opened. The command also reloads the file to the tester. See Loading and Unloading Patterns . In stand-alone mode, this command is disabled. See Stand-Alone PatternTool .
Open Last Burst	 Open Last Burst Opens the most recently executed pattern module, file, or set. The burst may have come from the test program, the Immediate window, or another tool. See Opening a File and Refreshing and Rebursting . In stand-alone mode, this command is disabled. See Stand-Alone PatternTool .

Close	Closes the current pattern file without exiting from PatternTool. If there are unsaved changes, you are prompted to save the changes. See Saving and Closing a File .
Save	 Ctrl+S Saves the current pattern file to its current name and folder. See Saving and Closing a File .
Save As	Saves the current pattern file to a new file you specify. You can specify the folder and filename. The new file becomes the current pattern file. If the new file already exists, you are asked if you want to overwrite the file. See Saving and Closing a File .
Import From Ascii	Compiles and opens a pattern from an ASCII source (.atp), using an interface similar to that of the Pattern Compiler. See Importing and Exporting ASCII Files .
Export To Ascii	Writes the current pattern file to an ASCII file (.atp), using the Pattern Reverse Compiler. See Importing and Exporting ASCII Files .
Exit	Exits from PatternTool. If there are unsaved changes to the current pattern file, you are prompted to save the changes.

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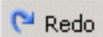
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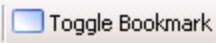
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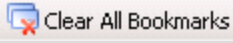
PatternTool Toolbar and Command Reference : Edit Menu Commands

Edit Menu Commands

Undo	 Undo Ctrl+Z
	Undoes the previous edit. The command indicates what the next Undo would affect: single or multiple cell edits, a shift, or a paste. See Undoing and Redoing Edits .
Redo	 Redo Ctrl+Y
	Redoes the previous undo. The command indicates what the next Redo would affect: single or multiple cell edits, a shift, or a paste. See Undoing and Redoing Edits .
Cut	 Ctrl+X
	Removes the data from the currently selected cells or vectors and puts it on the clipboard. See Cutting, Copying, and Pasting Cells and Inserting, Copying, and Deleting Vectors .
Copy	 Ctrl+C
	Copies the data from the currently selected cells or vectors onto the clipboard. See Cutting, Copying, and Pasting Cells and Inserting, Copying, and Deleting Vectors .
Paste	 Ctrl+V
	Overwrites the currently selected cells or vectors with the current clipboard contents. See Cutting, Copying, and Pasting Cells and Inserting, Copying, and Deleting Vectors .

Clear	Clears the contents of the currently selected cell or cells.
New Subroutine...	Moves or copies the selected vectors into a new subroutine. See Moving Vectors to a New Subroutine .
Delete	Deletes the currently selected item, from a pull-right menu: Pin from the Pin Explorer, Module from the Pattern Explorer, Vector or Comment from the main display.
Find	Ctrl+F Brings up the Find dialog box. See Using Find .
Replace	Ctrl+H Brings up the Replace dialog box. See Using Replace .
Mask Transition	Ctrl+H Brings up the mask Transitions dialog box, to specify which transitions are to be masked. See Masking Transitions .
Replicate...	Ctrl+R Repeats the selected cells the specified number of times toward the final vector. A window comes up to let you specify the number of times to replicate the selected data. See Replicating Values .
Replicate End of Module	Ctrl+M Repeats the selected cells to the end of the current module. See Replicating Values .
Replicate to End	Ctrl+E Repeats the selected cells to the final vector. See Replicating Values .
Shift...	Shifts the selected cells the specified number of vectors in the specified direction. See Shifting Values .
Complement	Changes pin states to their complement (for example, 0 to 1) in the selected cells. See Complementing and Swapping Values .
Swap...	Brings up the Swap dialog box. You can use the dialog box to perform a complement or other swap operations, plus an arbitrary swap that replaces specified pin states with other specified pin states. See Complementing and Swapping Values .
Toggle Bookmark	 Toggle Bookmark Ctrl+F2 Adds a bookmark to the currently selected vector, if it does not have one. Removes the bookmark if it has one. See Bookmarks .

Clear All Bookmarks



Shift+F2

Removes all bookmarks. See [Bookmarks](#).

Launch Editor

Launch the edit window for the currently selected cell.

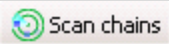
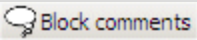
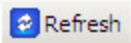
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PatternTool Toolbar and Command Reference : View Menu Commands

View Menu Commands

Pattern Explorer	Opens the Pattern Explorer panel. See Pattern Explorer.
Pin Explorer	Opens the Pin Explorer panel. See Pin Explorer.
Properties	Opens the Properties panel. See Viewing and Changing Pattern Properties and Viewing and Changing Pin Properties.
Message Display Panel	Opens the Message Display Panel. See About the Message Panel.
Show scan chains	 Scan chains Exposes or hides scan chains. The button is enabled only if the pattern contains scan chains. See Viewing and Editing Scan Chains.
Show block comments	 Block comments Exposes or hides block comments. The button is enabled only if the pattern contains block comments. See Viewing and Editing Block Comments.
Take Snapshot	Takes a snapshot of the current state of the tool. See Taking and Viewing Snapshots.
View Snapshot	Displays a snapshot that was previously taken. The pull-right menu lets you choose the Most Recent or Other, which displays a list of all snapshots in the test program folder. See Taking and Viewing Snapshots.
Restore Design View	Restores the tool layout to its original design.
Refresh	 Refresh F12

Refreshes the display after a pattern has been reburst from outside PatternTool. See Refreshing and Rebursting.

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PatternTool Toolbar and Command Reference : Insert Menu Commands

Insert Menu Commands

Pin	Inserts a new pin into the currently selected point in the main display or Pin Explorer. See Inserting, Deleting, and Modifying Pin Items.
Module...	Inserts a new module into the currently selected point in the Pattern Explorer. See Inserting, Renaming, and Deleting Modules.
Vectors...	Inserts new vectors above the currently selected vector in the main display area. See Inserting Vectors.
Block Comment	Inserts a block comment above the currently selected vector in the main display area. See Viewing and Editing Block Comments.
Copied or Cut Vector	Inserts previously copied or cut vectors above the currently selected vector in the main display area. See Cutting, Copying, and Pasting Vectors.

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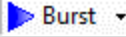
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PatternTool Toolbar and Command Reference : Debug Menu Commands

Debug Menu Commands

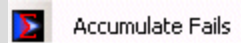
Burst	 Burst Ctrl+B
	Bursts the currently displayed pattern once. If the pattern has been modified, the button and command are Update and Burst. See Burst Options.
Burst Loop	 Burst Loop
	Bursts the currently displayed pattern in a loop. The status bar Burst Count indicates the number of times the pattern has been burst. You can use Halt to stop the loop. If the pattern has been modified, the button and command are Update and Burst Loop. See Burst Options.
Burst Until Pass	 Burst Until Pass
	Bursts the currently displayed pattern in a loop, until the pattern passes. The status bar Burst Count indicates the number of times the pattern has been burst. You can use Halt to stop the loop. If the pattern has been modified, the button and command are Update and Burst Until Pass. See Burst Options.
Burst Until Fail	 Burst Until Fail Ctrl+Shift+B

Bursts the currently displayed pattern in a loop, until the pattern fails. The status bar Burst Count indicates the number of times the pattern has been burst. You can use **Halt** to stop the loop.

If the pattern has been modified, the button and command are **Update** and **Burst Until Fail**.

See **Burst Options**.

Accumulate Fails

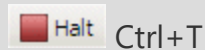


Repeatedly bursts the currently displayed pattern to accumulate the fails from the entire pattern. The status bar Burst count indicates the number of times the pattern has been burst. You can use **Halt** to stop the loop.

If the pattern has been modified, the button and command are **Update** and **Accumulate Fails**.

See **Accumulate Fails**.

Halt



Ctrl+T

Stops execution of a pattern. See **Burst Options**.

Set/Reset Capture Start

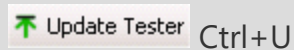
Ctrl+Q

Sets the point at which to start capturing HRAM. The default is vector 0. See **Setting Start and Stop**.

Set/Reset Capture Stop

Sets the point at which to start capturing HRAM. The default is when HRAM memory is full. See **Setting Start and Stop**.

Update Tester



Ctrl+U

Writes the current pattern file, including any edits, to tester memory. See **Updating the Tester**.

Update Tester (Force Reload)

Ctrl+Shift+U

Reloads the module and then updates tester memory. See **Updating the Tester**.

Unload Pattern

Ctrl+P

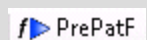
Unloads the current pattern. See **Loading and Unloading Patterns**.

Mask/Unmask

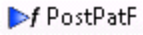
Ctrl+Shift+K

Masks or unmasks the currently selected pin. Masked pins appear in blue. See **Masking Pins and Vectors**.

PrePatF



Enables or disables the execution of the PrePatF function before each pattern burst. See **Burst Options**.

PostPatF	 PostPatF
	Enables or disables the execution of the PostPatF function after each pattern burst. See Burst Options.
HRAM Capture	Capture: Fails ▼
	Specifies what to capture: Fails, FirstModFail, or STV. See Setting What To Capture.
Learn ...	 Learn Ctrl+L
	Starts Learn mode, which lets you ignore pattern errors and update the pattern directly using HRAM. See Using Learn Mode.
Learn Mode Settings	Ctrl+D
...	Brings up a dialog box for specifying Learn mode settings. See Learn Mode Settings.

In stand-alone mode, all the Debug commands and corresponding buttons are disabled. See Stand-Alone PatternTool.

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PatternTool Toolbar and Command Reference : Tools Menu Commands

Tools Menu Commands

Waveform Display	Brings up the Waveform Display to show the results of executing the pattern. See Displaying Waveforms .
Waveform Display (Vector Mode)	Sends the selected block of data to Waveform Display for viewing in vector mode. See Viewing Vectors in Waveform Display .
Actual Mode	Brings up the Waveform Display to run Actual Mode on the selected cells. See Using Actual Mode .
HRAM Display	Brings up the HRAM Display tool. See About HRAM Display .
Protocol Studio	Brings up the Protocol Studio tool. See About Protocol Studio .
PatternTool (StandAlone)	Opens the current pattern in stand-alone mode. See Stand-Alone PatternTool .
Digital Channel	Opens the Digital Channel debug display for the currently selected pin or pin group.
PPMU	Opens the PPMU debug display for the currently selected pin.
Frequency Counter	Opens the Frequency Counter debug display for the currently selected pin or pin group.
Customize...	Not supported.
Options...	Not supported.

In stand-alone mode, none of the [Tools](#) commands are supported. See [Stand-Alone PatternTool](#).

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PatternTool Toolbar and Command Reference : Window Menu Commands

Window Menu Commands

Freeze View	Freeze the view so that columns to the left of the selected column remain static during scrolling. See Freezing and Unfreezing the View .
Unfreeze View	Unfreeze the view so that no columns remain static during scrolling. See Freezing and Unfreezing the View .



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Viewing Pattern Data

Viewing Pattern Data

This section describes aspects of viewing PatternTool data. The following topics are available:

- | | |
|---|-------------------------------------|
| • | PatternTool Display |
| • | Navigation |

For information on viewing scan chains, see [Viewing and Editing Scan Chains](#).

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Viewing Pattern Data : PatternTool Display

PatternTool Display

This section describes aspects of the PatternTool display and how to control it. The following topics are available:

- [Display Basics](#)
- [Refreshing and Rebursting](#)
- [Display Columns](#)
- [Displaying Pins, Groups, and Lists](#)
- [Viewing Block Comments](#)
- [Status Bar](#)
- [Taking and Viewing Snapshots](#)
- [Selecting a Site](#)
- [Freezing and Unfreezing the View](#)
- [Viewing Messages](#)

- [Viewing Vectors in Waveform Display](#)

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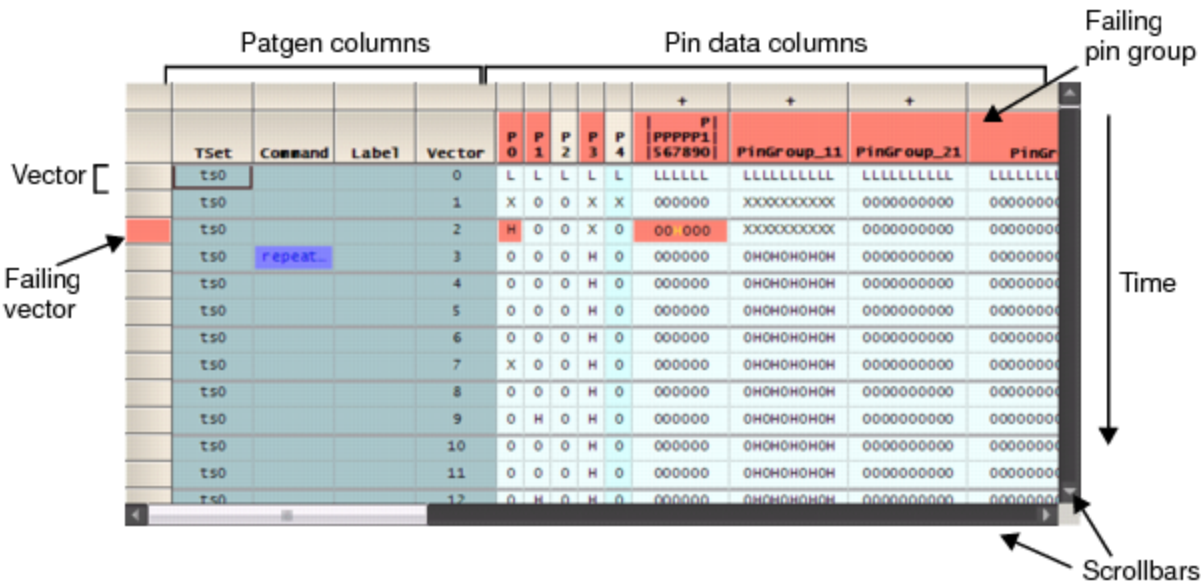
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Viewing Pattern Data : PatternTool Display : Display Basics

Display Basics

PatternTool displays the contents of the most recently burst pattern set, file, or module, or a pattern file that you have explicitly opened:



PatternTool Displaying a Pattern

The columns on the right side contain pin data. Each row represents a vector that will be applied by the PatGen hardware. Time increases in the downward direction. The vector number is a read-only field. The tool automatically increments the value from 0 in each module. In the case of scan vectors, the format is vectornumber.scanindex.

Horizontal and vertical scrollbars are presented as needed if there is more information than can be displayed within the viewing area. See Scrollbars.

Failing pin names and vectors have their topmost and leftmost columns colored pink. Cells at the intersection of a failing pin and vector are also in pink.

Major vector boundaries are represented as thick dark gray lines that run across the entire display. Module boundaries are represented as thick horizontal blue lines that run across the entire display,

between the last vector of a module and the first vector of the next module:

Vector boundary

Module boundary

	ts0			1908	0	0	0	0	0	000000
	ts0			1909	0	0	0	0	0	000000
	ts0			1910	0	0	0	0	0	000000
	ts0			0	0	0	0	0	0	000000
	ts0			1	0	0	0	0	0	000000
	ts0			2	H	0	0	0	0	000000
	ts0	repeat...		3	0	0	0	0	0	000000
	ts0			4	0	0	0	0	0	000000

PatternTool Vector and Module Boundaries

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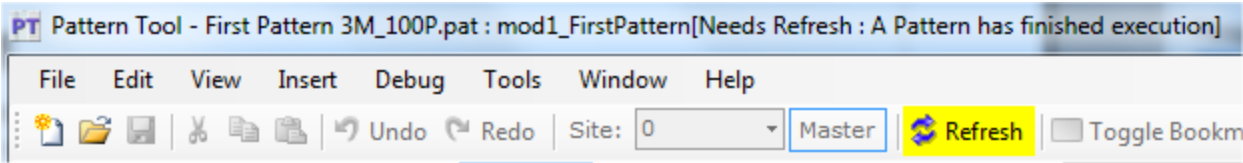
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Viewing Pattern Data : PatternTool Display : Refreshing and Rebursting

Refreshing and Rebursting

A pattern may be reburst outside PatternTool: for example, by another tool or by stepping through program code. In this case, the message “Needs Refresh” is displayed in the window title bar, along with the reason the refresh is needed. The Refresh button is also highlighted in yellow:



Needs Refresh

Click this button or use View>Refresh or F12 to display the results of the most recent burst. If you burst a pattern in PatternTool, the display is automatically refreshed.



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Viewing Pattern Data : PatternTool Display : Display Columns

Display Columns

The PatternTool display includes two types of columns:

- Patgen columns include Patgen information for each vector
- Pin data columns include the data for each pin, pin list, and pin group included in each vector

Patgen columns include:

TSet	The tset (time set) name or number for a vector. See Editing TSet Cells.
Command	PatGen microcode for a vector, including an opcode or control bits or both. See Editing Command Cells.
Label	The label for a vector. The format is type labelname: where type can be any valid label type. If type is omitted, the label is a local label. See Editing Label Cells.
Vector	The number for a vector starting with 0 for the first vector. You cannot edit cells in this column.

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Viewing Pattern Data : PatternTool Display : Displaying Pins, Groups, and Lists

Displaying Pins, Groups, and Lists

To display as much data as possible, pin names are displayed vertically at the top of the pin data column. The names of pin groups and lists (a pin list name is a parenthesized list of the pins) appear in the same space:

					Pin name					Pin list name		Pin group name	
	TSet	Command	Label	Vector	P 0	P 1	P 2	P 3	P 4	P 5	P 6	P 7	P 8
	ts0			0	L	L	L	L	L	LLLLL	LLLLLLLLL	LLLLLLLLL	LLLLLLLLL
	ts0			1	X	0	0	X	X	000000	XXXXXXXXX	000000000	000000000
	ts0	repeat...		2	H	0	0	X	0	000000	XXXXXXXXX	000000000	000000000
	ts0			3	0	0	0	H	0	000000	0H0H0H0H	000000000	000000000
	ts0			4	0	0	0	H	0	000000	0H0H0H0H	000000000	000000000
	ts0			5	0	0	0	H	0	000000	0H0H0H0H	000000000	000000000

Pin, Pin Group and Pin List Names in PatternTool

You can expand a pin group or list by clicking the plus sign (+) above its name or compress it by clicking the minus sign (-):

Pin list										Pin group									
(P5, P6, P7, P8,										PinGroup_11to20									
P 0	P 1	P 2	P 3	P 4	P 5	P 6	P 7	P 8	P 9	P 10	P 11	P 12	P 13	P 14	P 15	P 16	P 17	P 18	P 19
L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L	L
X	0	0	X	X	0	0	0	0	0	0	X	X	X	X	X	X	X	X	X
H	0	0	X	0	0	0	H	0	0	0	X	X	X	X	X	X	X	X	X
0	0	0	H	0	0	0	0	0	0	0	H	0	H	0	H	0	H	0	H
0	0	0	H	0	0	0	0	0	0	0	H	0	H	0	H	0	H	0	H

Pin List and Pin Group Expanded

If a collapsed pin group or list contains a pin with fails, the entire name cell is colored pink. The cell with the fail is colored pink, with the failing data bit in yellow, as shown above in the first figure. In expanded mode, only the pin with the fail is colored pink.

You can edit pin data in a group or list in either expanded or collapsed mode. If you double-click a cell in collapsed mode, a special window comes up for editing bus data. See Pin Group (Bus) Data Edit Window.

Pins with the setup 2X use two pin columns to display their data:

	d a t a i n 2 x	d a t a i o 2 x	d a t a i n	d a t a i o	
Vector					Comment
0	00	XX	0	L	
1	01	XX	0	L	
2	10	XX	1	H	
3	11	XX	1	H	
4	00	LL	0	L	
5	01	LH	1	H	

2X Pins Displayed in Two Columns

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Viewing Pattern Data : PatternTool Display : Viewing Block Comments

Viewing Block Comments

The IG-XL pattern compiler supports block comments, which appear between vectors:

```
> - 0 1 00000100 LLLLLHLL ;
/* Multi-line block comment      */
/*      between vectors      */
> - 0 1 00000101 LLLLLHLH ;
```

PatternTool lets you view, add, and edit block comments.

 **Block comments** Use this button or the View>Show Block Comments command to toggle between showing and hiding block comments.

The button is enabled only if the pattern displayed contains block comments.

For details on how the pattern compiler treats block comments, see [Comments in Pattern Files](#) in the pattern language reference. For example, the Pattern Compiler may combine or move certain block comments so that they are different from the way they appear in the .atp file.

In addition to block comments, the .atp pattern source file can also have Start Comments and a Final Comment. See [Comments in Pattern Files](#). PatternTool does not display Start or Final Comments. Note that any block comment that appears after the final vector in a module is assumed to be part of the Start Comment for the next module. PatternTool does not display such comments.

A block comment is associated with a specific vector, and that the block comment appears before its vector.

When displaying block comments is enabled, you can use the Goto navigation control to go to the previous or next block comment. See [Goto](#) in the section on navigation.

Note that end-of-line comments (which use // in the source .atp file) are always visible, regardless of the Block comments setting. End-of-Line comments appear in a separate Comment column.

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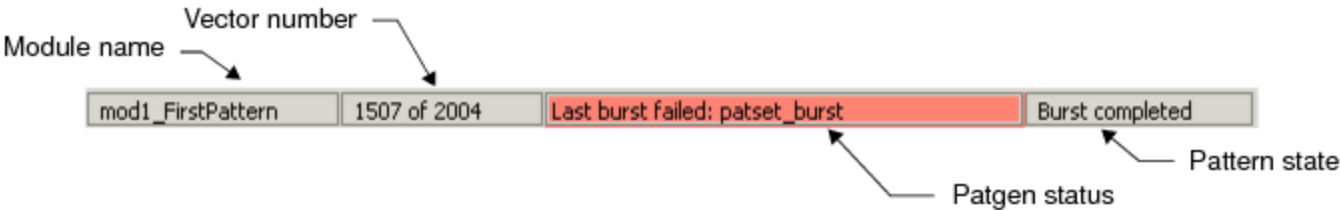
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Viewing Pattern Data : PatternTool Display : Status Bar

Status Bar

The PatternTool status bar looks like this:



PatternTool Status Bar

The status bar displays the following information:

Module name	The name of the module in which the currently selected cell is located.
Vector number	The vector number of the currently selected cell, out of the total number of vectors.
Patgen status	The most recently burst pattern and the pass/fail result of the burst.
Pattern state	The state of the pattern currently displayed. The pattern is open or the reburst completed. If you are bursting until fail, this item shows the Burst count, which is the number of times the pattern looped until a failure occurred or you halted the burst.

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Viewing Pattern Data : PatternTool Display : Taking and Viewing Snapshots

Taking and Viewing Snapshots

PatternTool lets you take snapshots of the current state of the tool. Later you can view snapshots for a visual comparison of tool states.

The [command](#) View>Take Snapshot takes a snapshot of the PatternTool window. The View>View Snapshot command lets you view the snapshots. It has these options:

Most Recent	Displays the most recent snapshot.
Other	Brings up a browser to let you select the snapshot to view.

The display uses the default Windows image viewer.

By default, a snapshot is saved to the test program folder. If the default folder is read-only, it is saved to the tmp folder. You are not prompted to specify a folder. A snapshot is saved as a JPEG file with a filename of this form:

Pattern Tool_YMMMDDHHMMSS.jpg

This means that snapshots in the browser will be in chronological order, with the most recent snapshot at the bottom of the list.

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Viewing Pattern Data : PatternTool Display : Selecting a Site

Selecting a Site

Site: 0 ▾ Master

PatternTool shows the data for one site at a time. Use the site selector to specify a site.

If you click Master, PatternTool uses the master site as it is defined in DataTool. The site number displays the current master site, but it is grayed out because you cannot change it. If the master site changes, PatternTool indicates that you need to refresh the display.

If you disable Master, you can manually select the site to be displayed. If you change the site, PatternTool autorefreshes to display the data from the new site.

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Viewing Pattern Data : PatternTool Display : Freezing and Unfreezing the View

Freezing and Unfreezing the View

When you scroll the display, you may want to keep some columns, such as patgen data, in view. You can do this by freezing the view.

By default, patgen columns are kept static at the left of the display area. This is indicated by the orange line to the right of the Vector column:

Static line

TSet	Command	Label	Vector	PingGroup_51to65	PingGro	PingGroup_71	PingGro	PingGroup_86to99	Comment
ts0			0	LLLLLLLLLLLLLLLL	LL L...	LLLLLLLLLL	LLLLL	LLLLLLLLLLLLLLLL	
ts0			1	XXXXXXXXXXXXXXXX	00 0...	0000000000	00000	00000000000000	
ts0	repeat 1000		2	XXXXXXXXXXXXXXXX	00 0...	0000000000	00000	00000000000000	
ts0			3	LHLHLHLHLHLHLHL	00H00	0000000000	00000	00000000000000	
ts0			4	LHLHLHLHLHLHLHL	00H00	0000000000	00000	00000000000000	
ts0			5	LHLHLHLHLHLHLHL	00H00	0000000000	00000	00000000000000	
ts0			6	LHLHLHLHLHLHLHL	00H00	0000000000	00000	00000000000000	
ts0			7	LHLHLHLHLHLHLHL	00H00	0000000000	00000	00000000000000	
ts0			8	LHLHLHLHLHLHLHL	00H00	0000000000	00000	00000000000000	
ts0			9	LHLHLHLHLHLHLHL	00H00	0000000000	00000	00000000000000	
ts0			10	LHLHLHLHLHLHLHL	00H00	0000000000	00000	00000000000000	
ts0			11	LHLHLHLHLHLHLHL	00H00	0000000000	00000	00000000000000	
ts0			12	LHLHLHLHLHLHLHL	00H00	0000000000	00000	00000000000000	
ts0			13	LHLHLHLHLHLHLHL	00H00	0000000000	00000	00000000000000	
ts0			14	LHLHLHLHLHLHLHL	00H00	0000000000	00000	00000000000000	
ts0			15	LHLHLHLHLHLHLHL	00H00	0000000000	00000	00000000000000	
ts0			16	LHLHLHLHLHLHLHL	00H00	0000000000	00000	00000000000000	
ts0			17	LHLHLHLHLHLHLHL	00H00	0000000000	00000	00000000000000	
ts0			18	LHLHLHLHLHLHLHL	00H00	0000000000	00000	00000000000000	
ts0			19	LHLHLHLHLHLHLHL	00H00	0000000000	00000	00000000000000	
ts0			20	LHLHLHLHLHLHLHL	00H00	0000000000	00000	00000000000000	
ts0			21	LHLHLHLHLHLHLHL	00H00	0000000000	00000	00000000000000	

Static Line Indicating Columns to the Left Are Frozen

You can change where the view is frozen. Select the column to the left of where you want the view frozen. You must select the entire column by clicking the column header. Then right-click and select Freeze View from the context menu. You can also use the Window>Freeze View command. The orange line is moved to the new location to the left of the selected column. All columns to the left of the line remain static during scrolling.

If you do not want the view frozen, use Window>Unfreeze View. The command is available on the right-click context menu as long as you do not have an entire column selected. No columns remain static during scrolling.

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Viewing Pattern Data : [PatternTool Display](#) : Viewing Messages

Viewing Messages

Messages that do not require an immediate user response are written to Message Panel. For more information, see [About the Message Panel](#).

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Viewing Pattern Data : PatternTool Display : Viewing Vectors in Waveform Display

Viewing Vectors in Waveform Display

PatternTool displays a pattern file in vector mode. You may want to see a block of data in waveform mode. You can select a contiguous block of vectors and pins in the PatternTool display grid. Then select Tools>Waveform Display (Vector Mode), or select the command from the context menu. This command sends the block of data to Waveform Display in vector mode, which displays the data as a waveform. You cannot burst or perform actual mode in vector mode.

By default, Waveform Display needs to burst a pattern to display the cycles. Going from PatternTool in vector mode is particularly useful when you are working on the simulator, where bursting a large pattern can take too long.

For more information, see About Waveform Display.



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Viewing Pattern Data : Navigation

Navigation

PatternTool provides the following ways to navigate to specific points in the display area:

- Goto
- Bookmarks
- Viewstrips
- Scrollbars

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Viewing Pattern Data : Navigation : Goto

Goto

The Goto control on the toolbar lets you go to a specific item in the display:



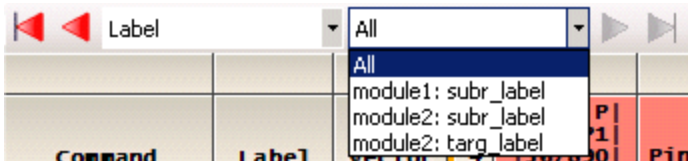
Goto Control

Use the drop-down list to select the type of information you want to go to: for example, a failure or a vector number. To specify which item of that type to go to (for example, which vector number), the search field to the right is enabled for you to enter or select the value:



Goto Control with Text Box

If the type has a defined number of instances, such as labels, a drop-down list is enabled:



Goto Control with Drop-Down List

You can select a specific item to go to, or you can select All to navigate through all instances of an item.

The red arrows let you go (from left to right) to the first, the previous, the next, or the last instance of the specified item. If you are going to a specific item, such as to a specific vector number or label, only the Next button is enabled; it takes you to the specified item.

The following are the types in the drop-down list, with some notes:

Fail

Failure navigation performs scrolling in the vector axis only. It does not affect the scroll location in the pin axis. You may need to scroll horizontally to see the failure.

Vector	Navigate to the vector identified by the number you enter here.
Label, Global Label Local Label	Navigate to any label or a global or local label. The drop-down lists all labels of that type, plus All to navigate to all labels of that type. If you right-click in a Command cell that has "call <label>," the context menu has a GoTo Label command that takes you to the vector with the label.
Scan Vector	Navigates to vectors with the scan opcode, not to scan chains. See Viewing and Editing Scan Chains.
Pin/Group Name	Navigate to the pin or group name you specify. Matching is case-insensitive. Wildcard characters * and ? are supported since you may not always want to match an exact pin name. When you use one of the wildcard characters, the Next and Previous buttons are available. If the target pin is in a collapsed bus, the bus is expanded.
Text	Enter a string, such as a label name or an opcode. Wildcard characters are not supported.
Data Sequence	Navigate to the sequence of data values that you enter into the search field. For example, you can enter "1H" (case is not significant). It searches pin-by-pin along the vertical pin axis, to find the next or previous occurrence of this sequence.
Bookmark	Navigate to a bookmark. See Bookmarks.
Block Comments	Navigate to the next or previous block comment containing the text that you enter in the search field. By default, the search field contains "*", which matches all block comments. See Viewing and Editing Block Comments.

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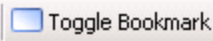
<

>

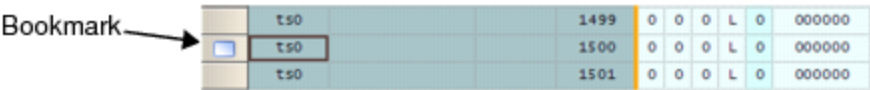
Viewing Pattern Data : Navigation : Bookmarks

Bookmarks

Bookmarks are a special category of Goto navigation.

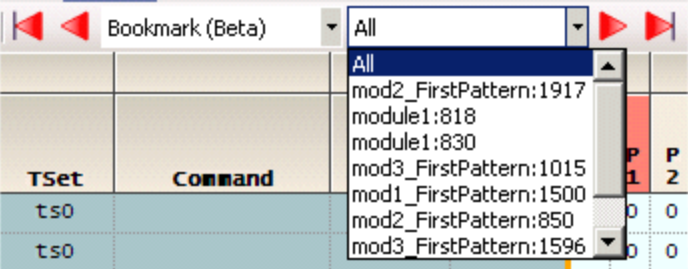
 You can put a bookmark at your current location using this button, the Edit>Toggle Bookmark command, or Ctrl+F2.

A bookmark looks like this:



PatternTool Bookmark

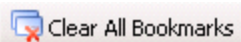
You can then select the Bookmark type in the Goto control:



Selecting a Bookmark in the Goto Control

The drop-down lists all bookmarks in the current display, including module name and vector number. It also includes All to let you navigate through all bookmarks.

You can clear one bookmark by selecting its vector and using the Toggle Bookmark button or command.

 To clear all bookmarks, use this button, the Edit>Clear All Bookmarks command, or Shift+F2.

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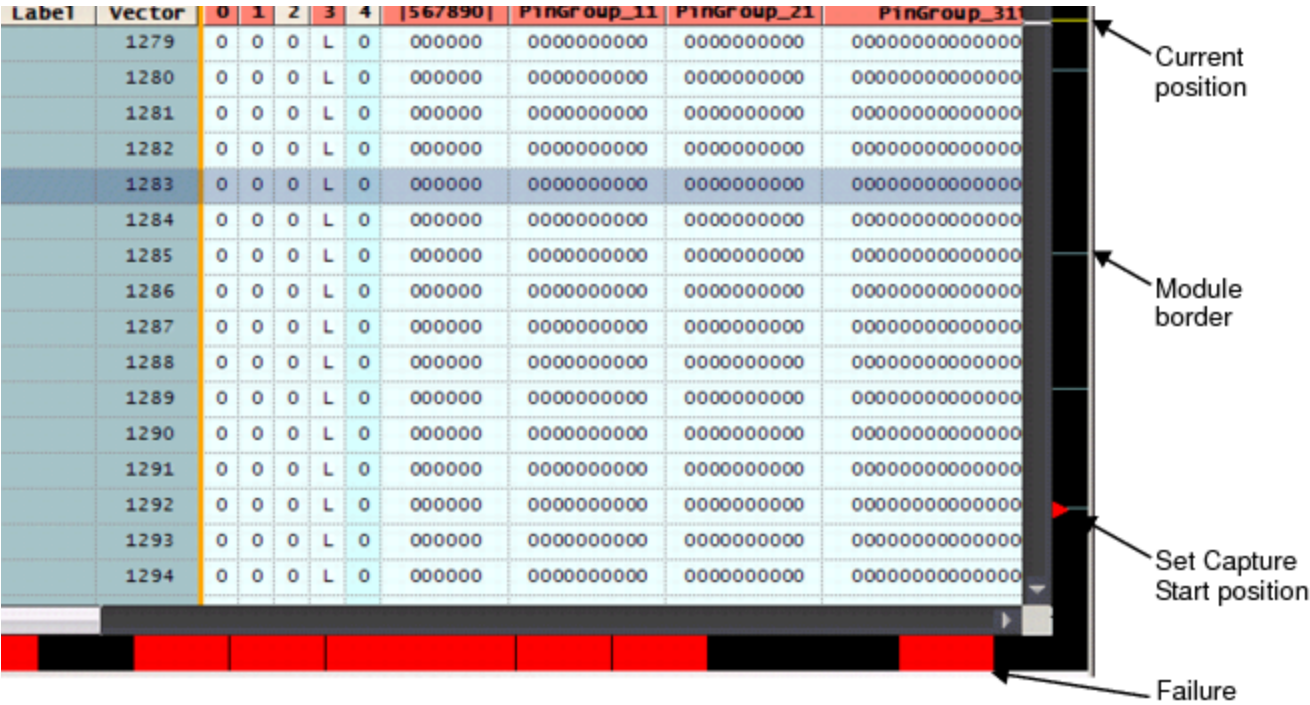
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Viewing Pattern Data : Navigation : Viewstrips

Viewstrips

If the data to be displayed extends beyond what is visible in the display area, you see a horizontal viewstrip, a vertical viewstrip, or both:



Viewstrips in the PatternTool Display

Viewstrips give an overall view of data, specifically of all captured failures across the horizontal time scale and the vertical vectors. Viewstrips show the relative positions of failures in relation to the entire span of data.

You can double-click at any point in a viewstrip to navigate to that position in the data. Clicking in the horizontal viewstrip scrolls only on the horizontal pin axis. Clicking in the vertical viewstrip scrolls only on the vertical vector axis.

The following colors are used to indicate items in the viewstrip:

- A red band indicates a failure at that location. If you hover over a failure on the horizontal viewstrip, a tooltip displays the fail pin name. If you hover over the vertical viewstrip, the vector number is displayed.
- The yellow line on the vertical strip indicates the current selected position.
- A blue line on the vertical strip indicates the border between two modules.
- A red triangle on the vertical strip indicates the Set Capture Start position.

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Viewing Pattern Data : Navigation : Scrollbars

Scrollbars

If the data to be displayed extends beyond the display area, you see a horizontal scrollbar, a vertical scrollbar, or both at the right and bottom of the tool.

You can use the arrows at each end of the scrollbar to move smoothly in that direction. You can left-click the scrollbar to jump a small distance in that direction. If you right-click a scrollbar, the context menu gives you more options:

- Scroll here, that is, to the point at which you clicked on the scrollbar.
- Scroll all the way to the top or bottom or left edge or right edge.
- Move one page’s worth up or down or to the right or left.

When you are scrolling, you may want to keep certain columns, such as the patgen data, in view. See [Freezing and Unfreezing the View](#).

